CMPS453 Cloud Computing Networking

Ayman AboElHassan, PhD
Assistant Professor
ayman.abo.elmaaty@eng.cu.edu.eg

Overlapping Topic

1. Software Engineering
2. Advanced Programming Techniques
3. Networks
4. Web
5. …

Our Focus

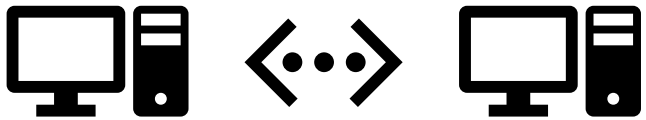
1. Networking basics
2. Networks inside Data Centers
3. Virtual Networks

What is a Network?

A group of connected devices that communicate with each other to share date and resources

Network Properties

1. Devices
2. Group
3. Communicate
4. Share data/resources



Communication Model

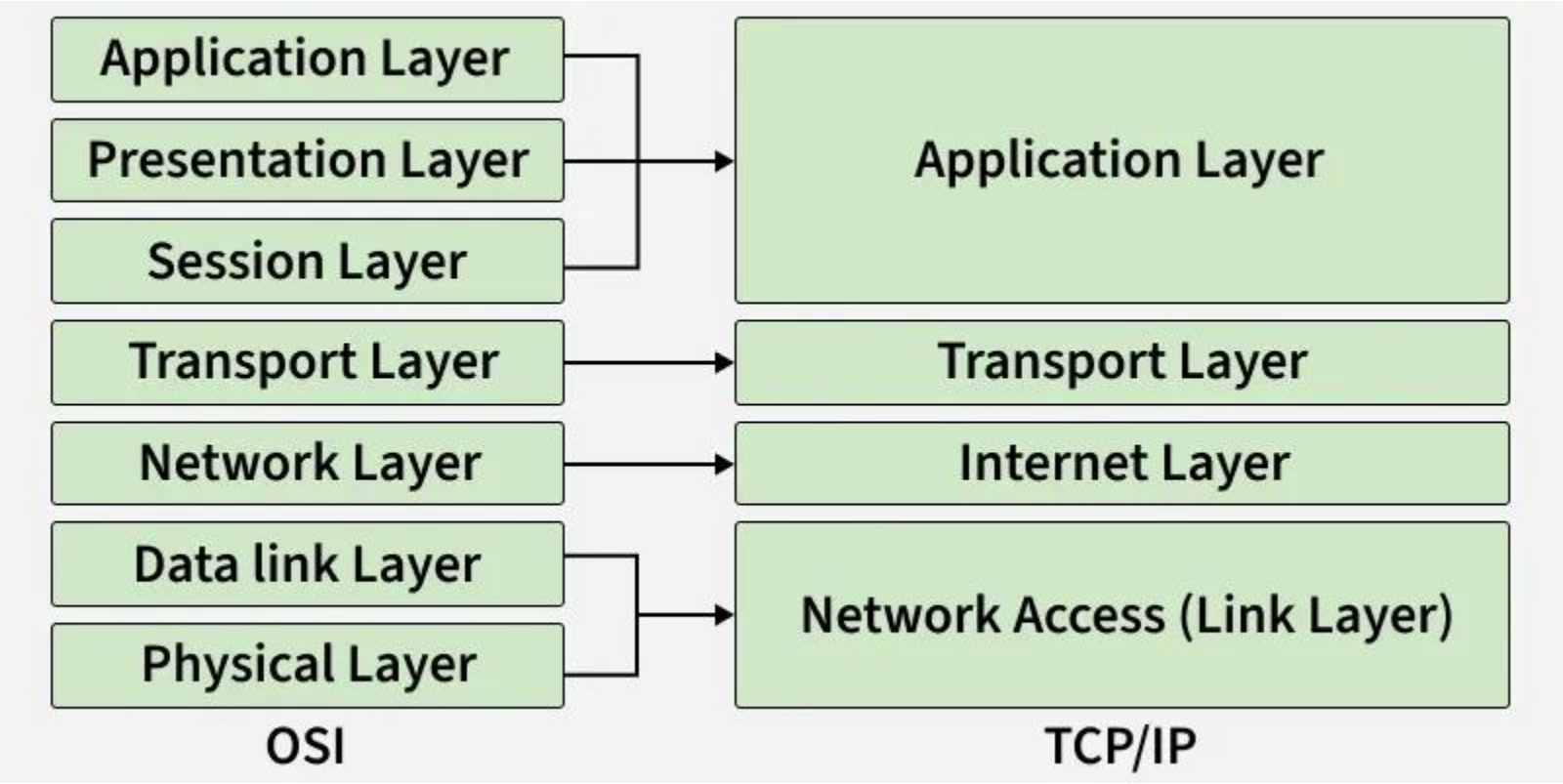
Communication Model

OSI Model

- Standardization (ISO)
- Defining role per layer

TCP/IP Model

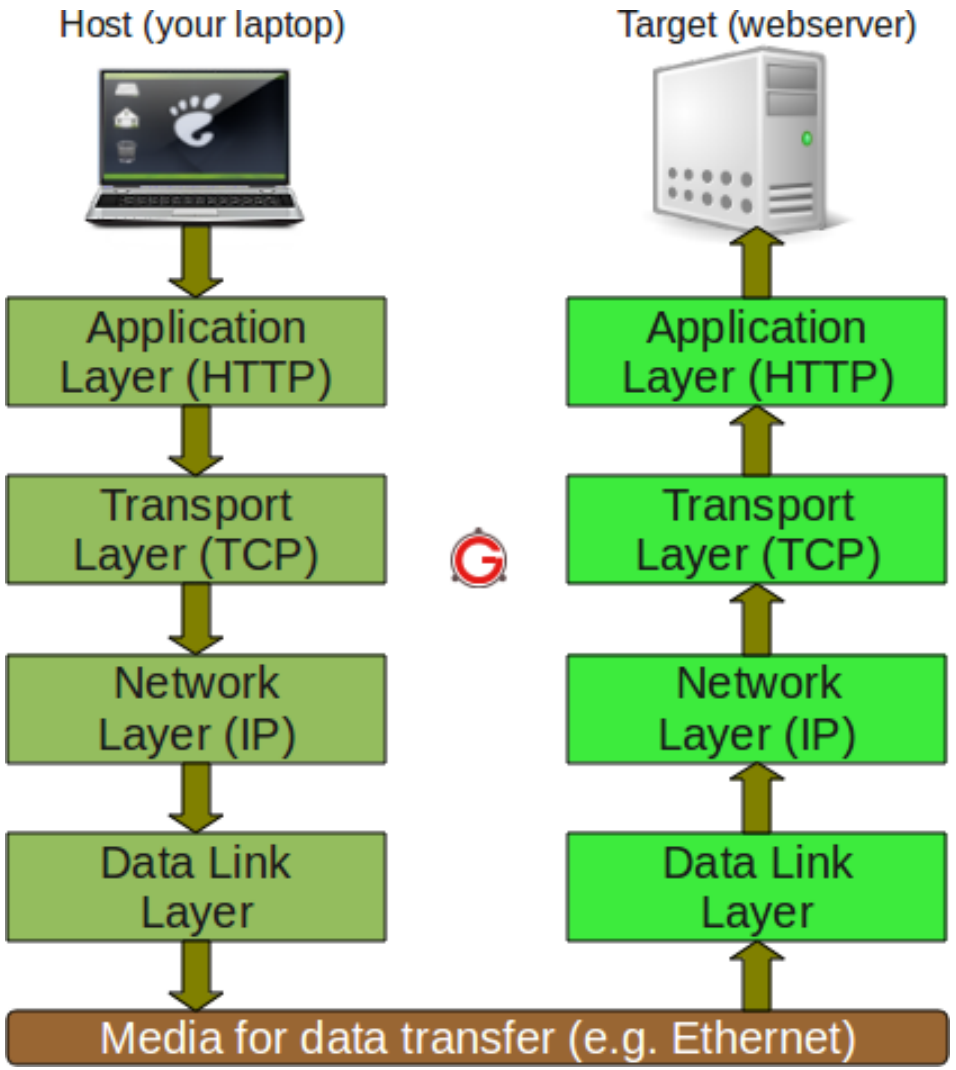
- Simpler structure
- Protocol-Driven (Practical)



Communication Model Example

Load “www.google.com” page

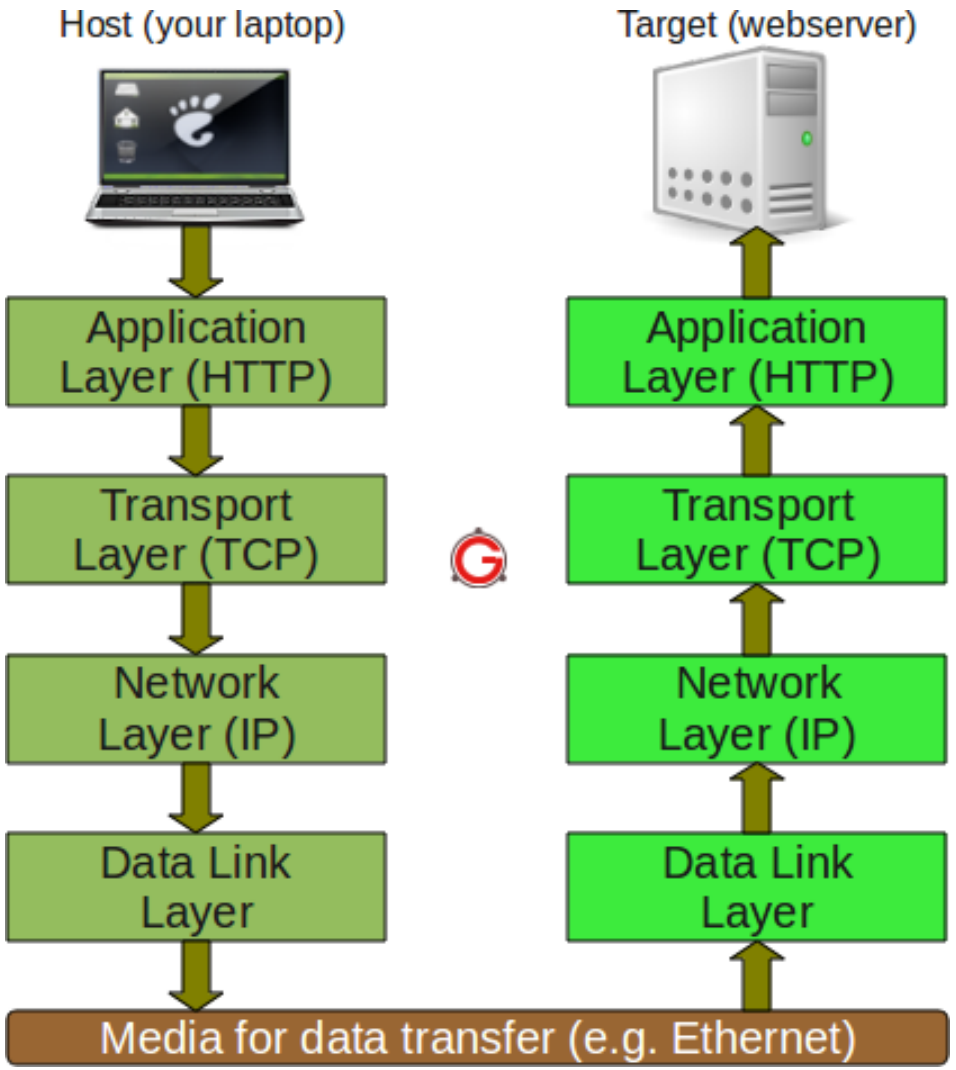
1. Browser sends HTTP request
2. Render the HTML page received



Communication Model Example

Load “www.google.com” page

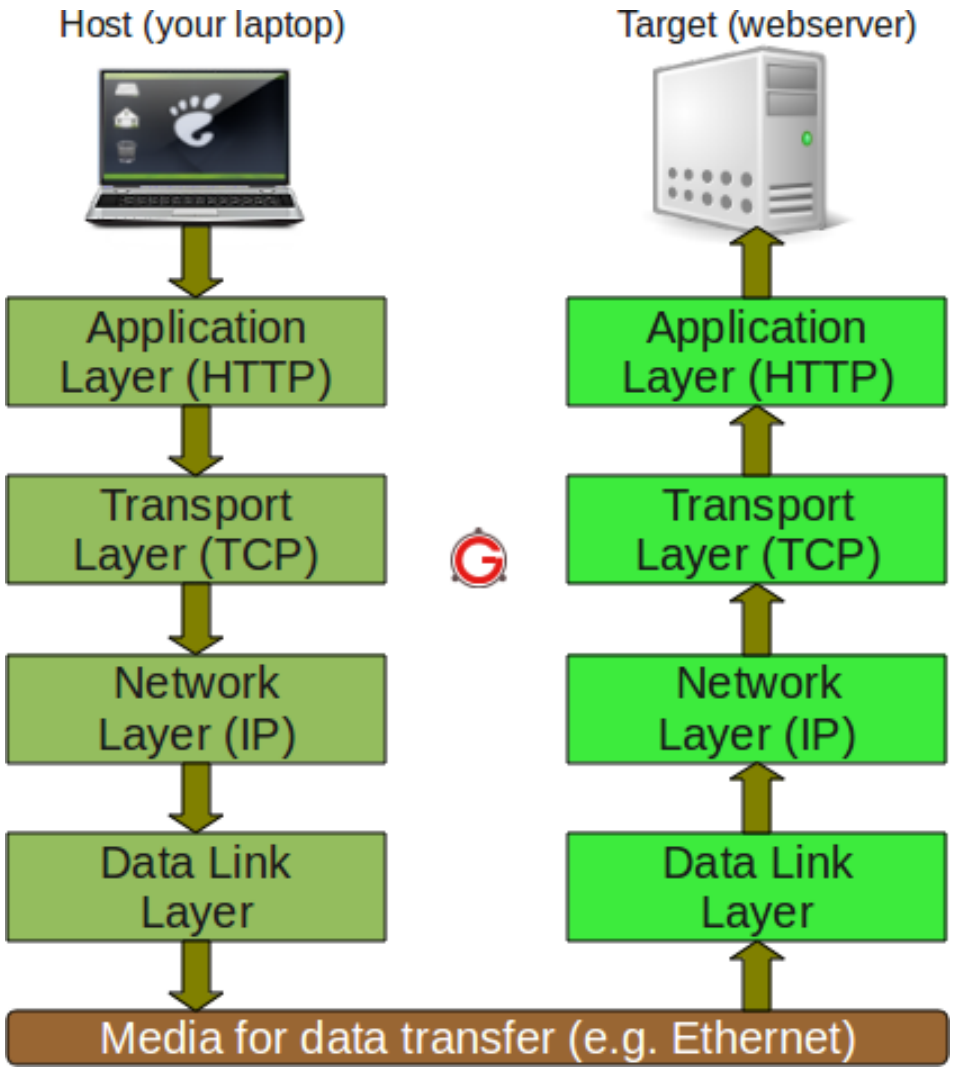
1. Browser sends HTTP request
   1. Ask DNS for target IP
   2. Open TCP connection (Handshake)
   3. Send TCP message of the request
   4. Wait for Ack (message received)
   5. Wait for response
   6. Send response to Browser
2. Render the HTML page received



Communication Model Example

Load “www.google.com” page

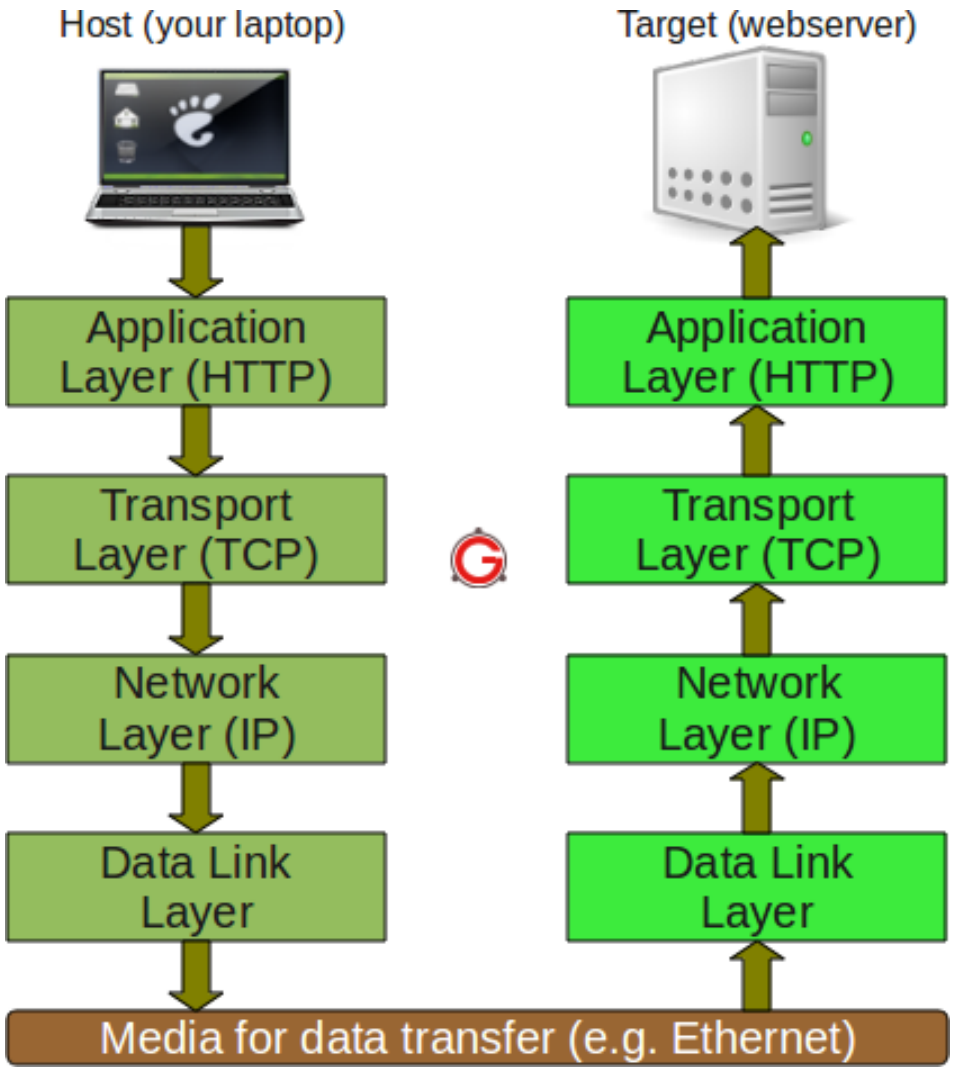
1. Browser sends HTTP request
   1. Ask DNS for target IP
   2. Open TCP connection (Handshake)
      1. Send IP packet with TCP “Sync” data to destination
      2. Wait for an IP packet with TCP “Ack” data from the destination
   3. Send TCP message of the request
      1. Send a set of IP packets with sequence ID to the destination
   4. Wait for Ack (message received)
      1. Wait for IP packet with TCP “Ack” data for each IP message sent before
   5. Wait for response
      1. Wait for a set of IP packets with a new sequence ID from the destination
   6. Send response to Browser
2. Render the HTML page received



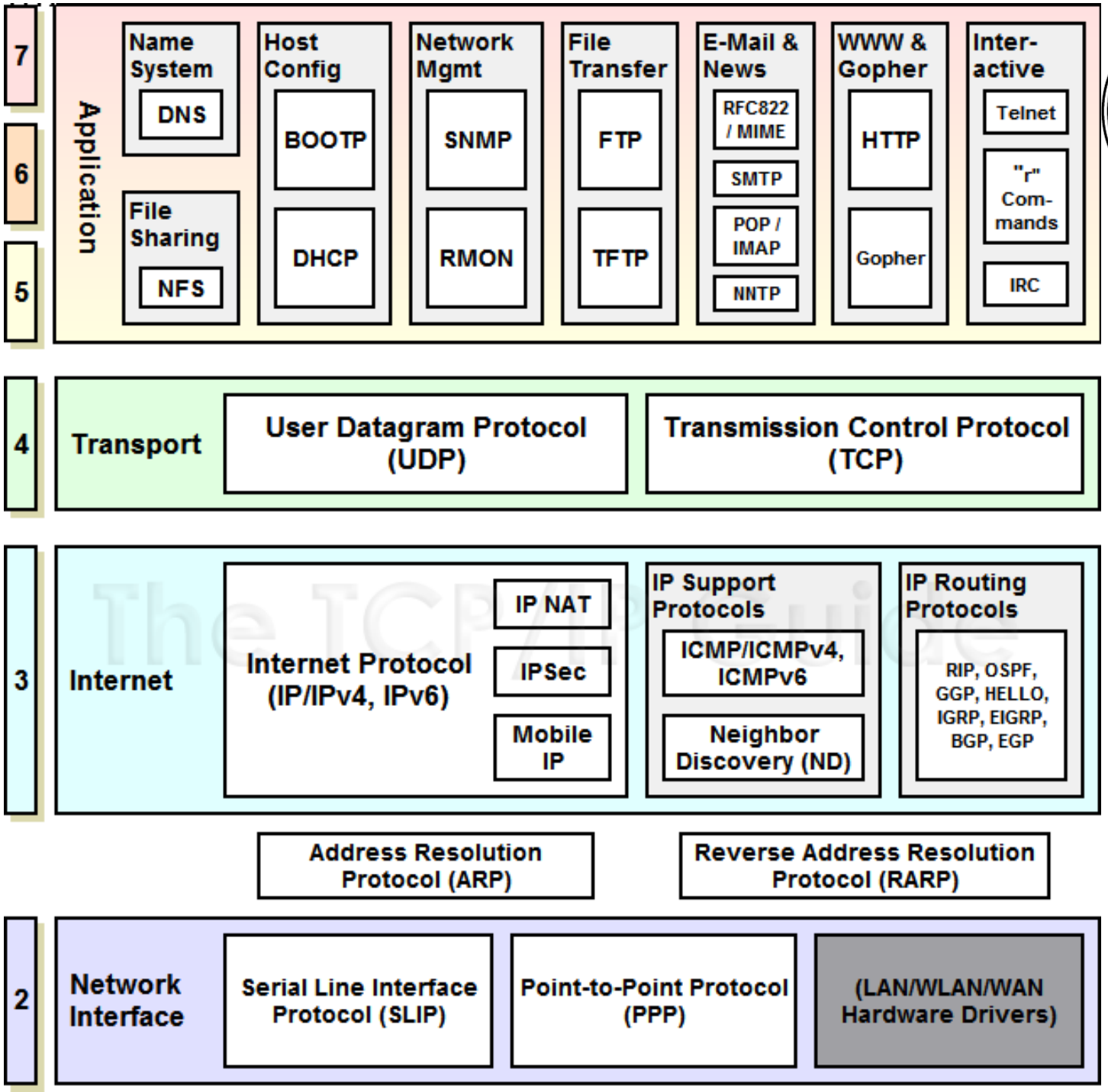
Communication Model Example

Load “www.google.com” page

1. Browser sends HTTP request
    1. Ask DNS for target IP
    2. Open TCP connection (Handshake)
        1. Send IP packet with TCP “Sync” data to destination
            1. System Call for NIC access to send data
        2. Wait for an IP packet with TCP “Ack” data from the destination
            1. System Call for NIC access to receive data
    3. Send TCP message of the request
        1. Send a set of IP packets with sequence ID to the destination
            1. System Call for NIC access to send data
    4. Wait for Ack (message received)
        1. Wait for IP packet with TCP “Ack” data for each IP message sent before
            1. System Call for NIC access to receive data
    5. Wait for response
        1. Wait for a set of IP packets with a new sequence ID from the destination
            1. System Call for NIC access to receive data
    6. Send response to Browser
2. Render the HTML page received



Communication Model



5. Application Layer

How to provide network services to end-user applications?

- Communication Protocols ➔ Used by APIs
  - HTTP
  - SMTP
  - SSH
  - FTP
  - NFS
  - WebSocket
  - …

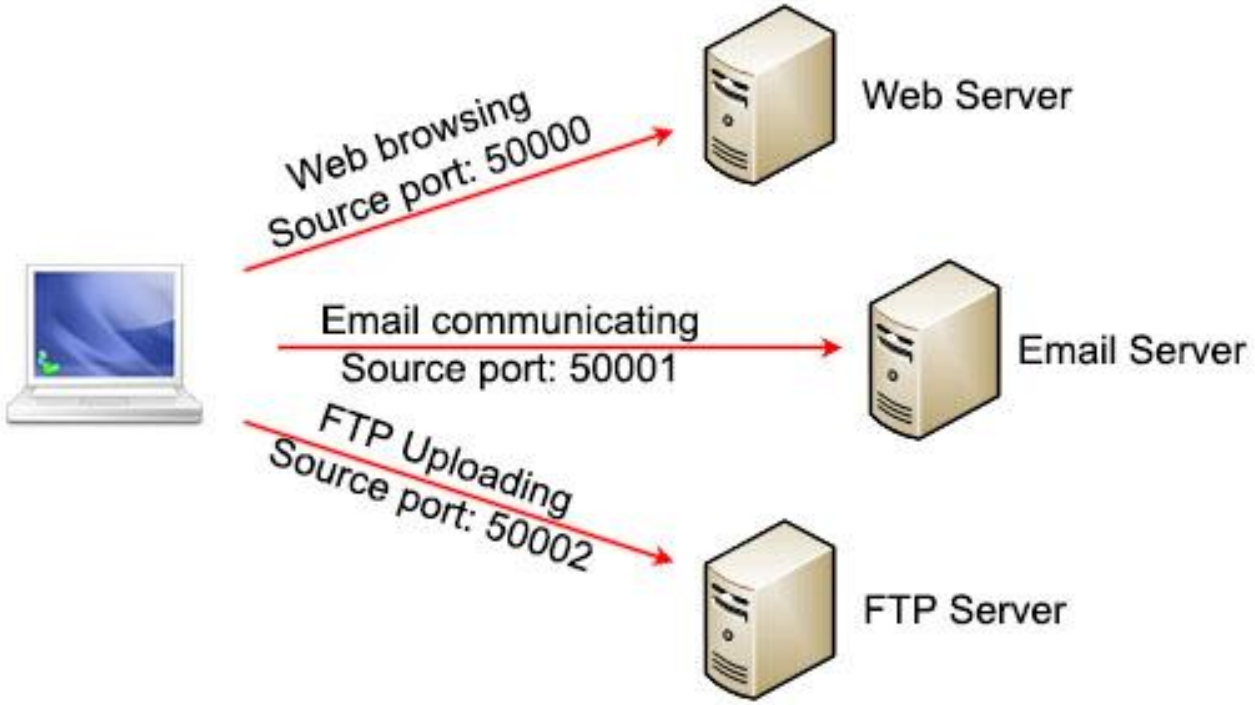
We will go back to it Later

4. Transport Layer

How to provide end-to-end communication?

- TCP/UDP & ports ➔ Which app sending/receiving the data?

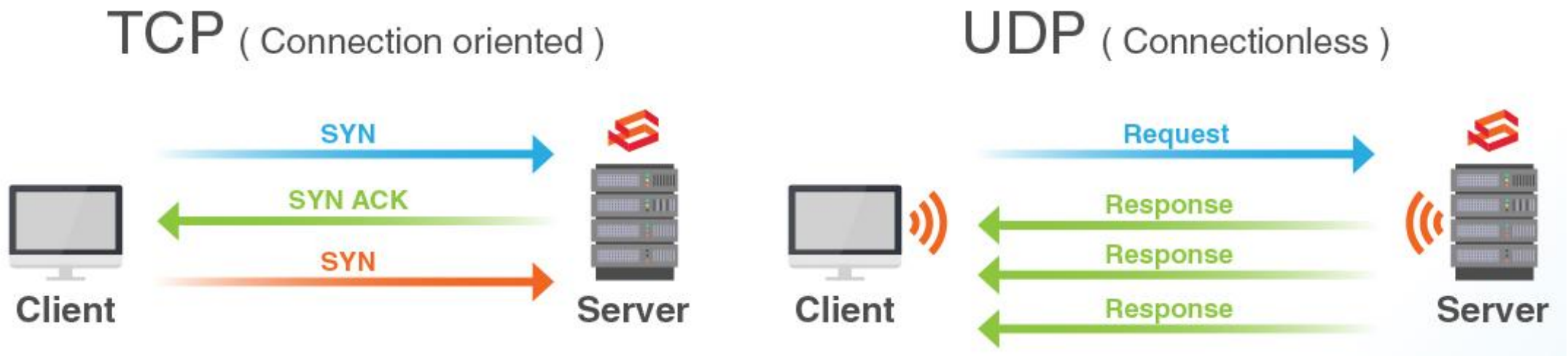
How to get your opened ports?



4. Transport Layer

How to provide end-to-end communication?

- TCP/UDP & ports ➔ Reliability vs Efficiency

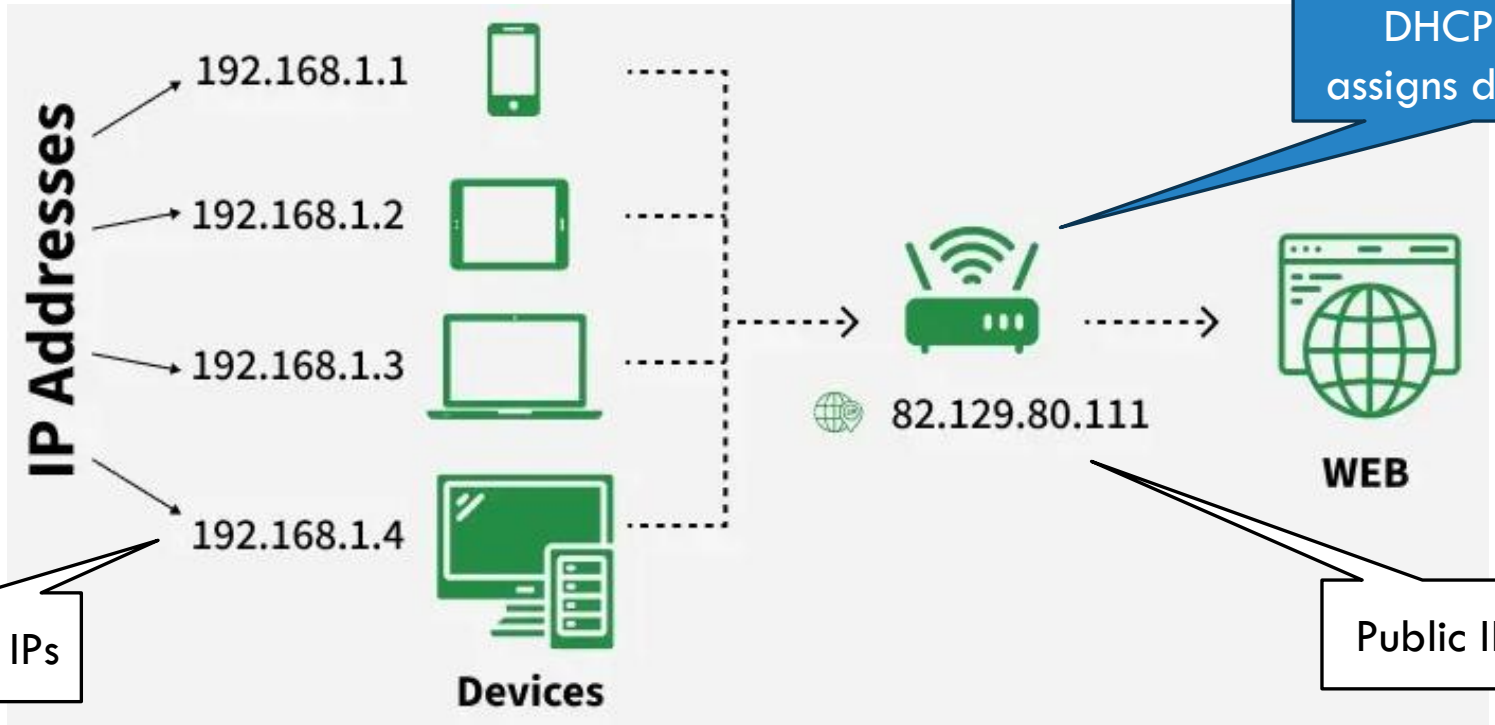


3. Network Layer

How data reaches the destination?

- IP addresses and routing

How to get your private/public IP address?



DHCP Server assigns dynamic IPs

Generated by the Router

Private IPs

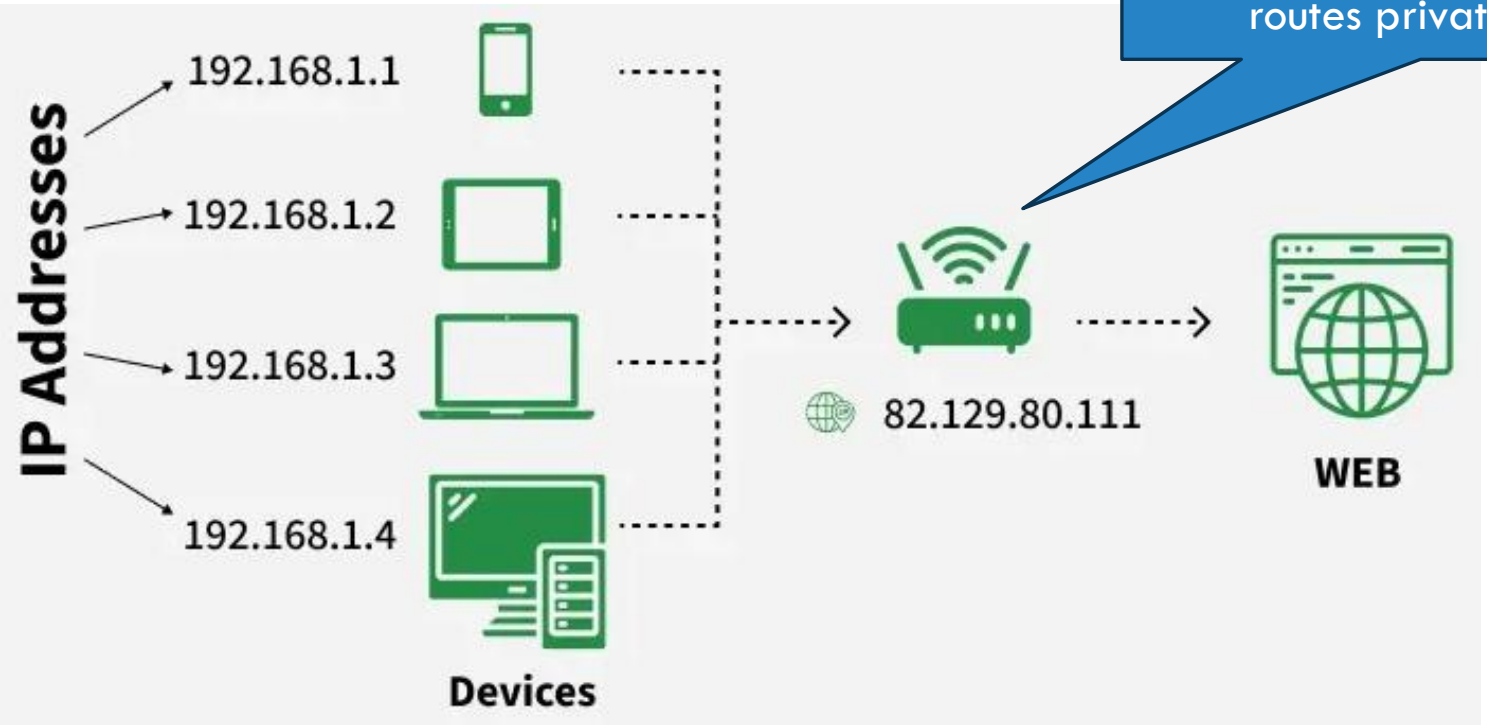
Public IPs

Generated by ISP

3. Network Layer

How data reaches the destination?

- IP addresses and routing

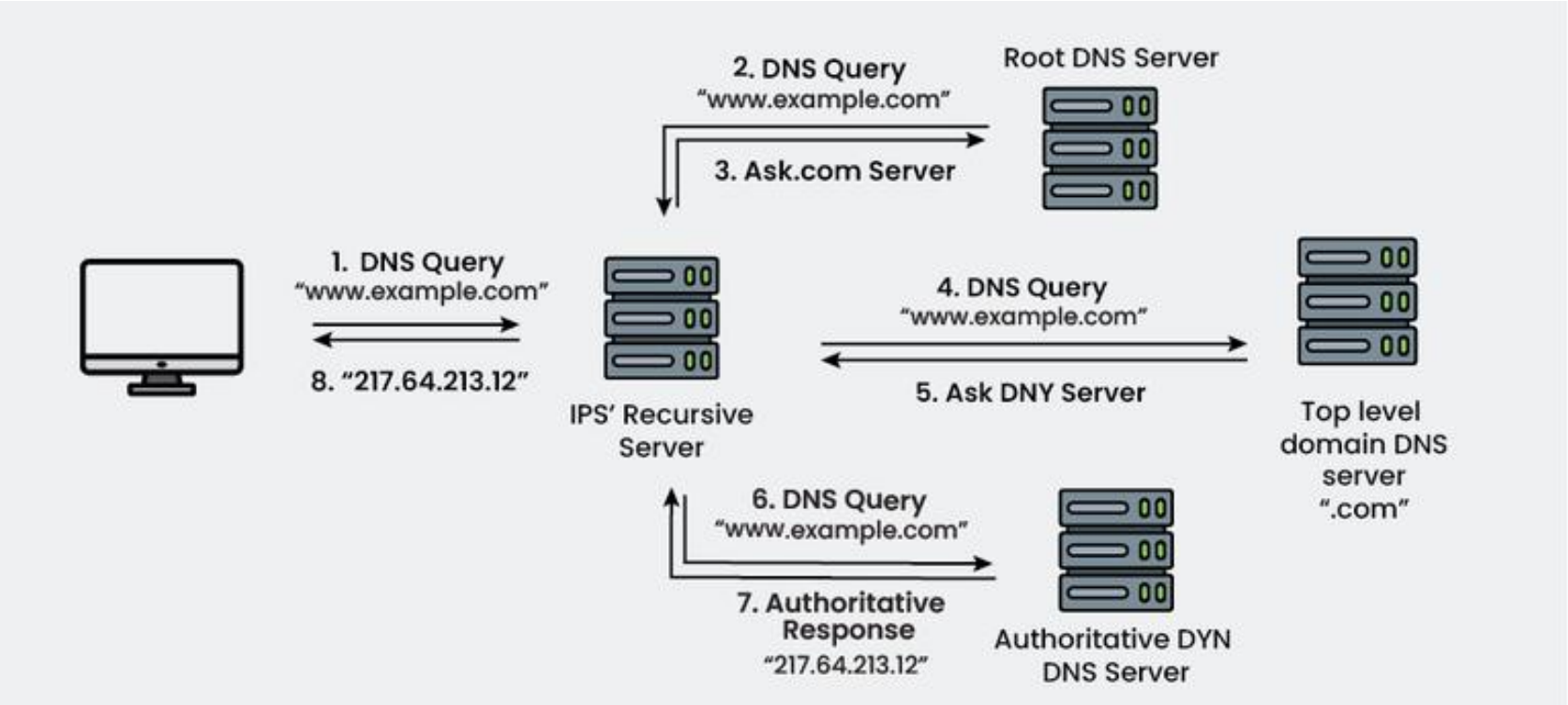


NAT (Network Address Translator) routes private/public IPs

3. Network Layer

How data reaches the destination?

- IP addresses and routing ➔ where is www.google.com?

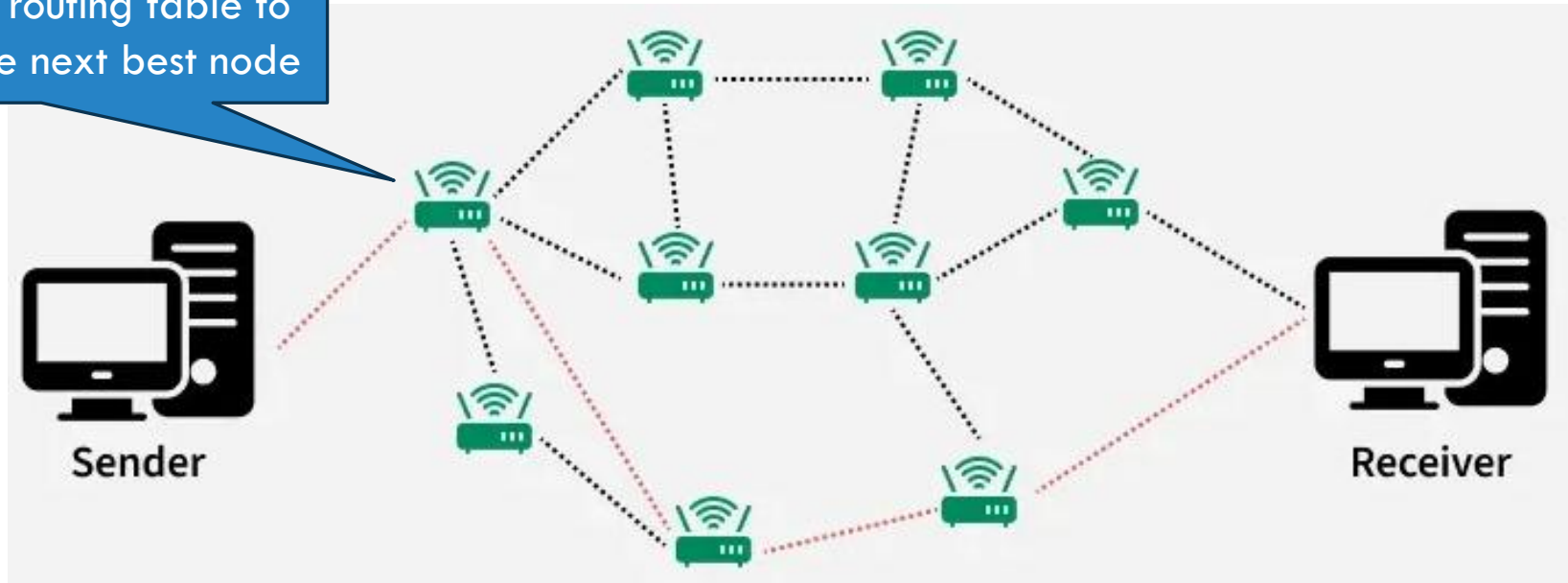


3. Network Layer

How data reaches the destination?

- IP addresses and routing ➔ How to reach 142.251.142.206?

Each router has a routing table to pass the msg to the next best node



2. Data Layer

How is data transferred?

- IEEE 802.3 → Ethernet
- IEEE 802.11 → Wi-Fi
- IEEE 802.15.1 → Bluetooth
- IEEE 802.15.4 → Zigbee
- 3GPP LTE R8 → 3G
- 3GPP LTE R10 → 4G
- 3GPP LTE R13 → 4.5G

2. Data Layer

How is data transferred?

- IEEE 802.3 ➔ Ethernet

Bit-by-Bit Transmission

Notification for Receiver Synchronization

Start Frame "10101011"

MAC Address

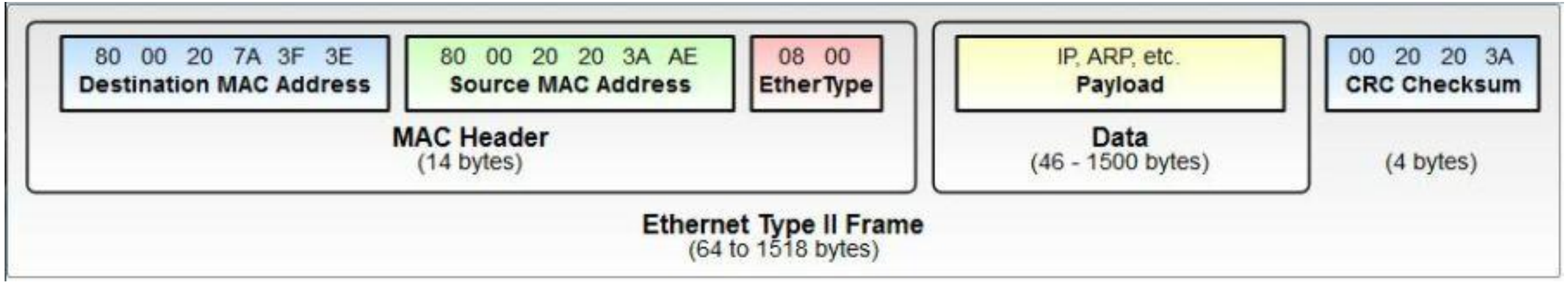
Payload size

32-bits Hash code

2. Data Layer

How is data transferred?

- IEEE 802.3 ➔ Ethernet

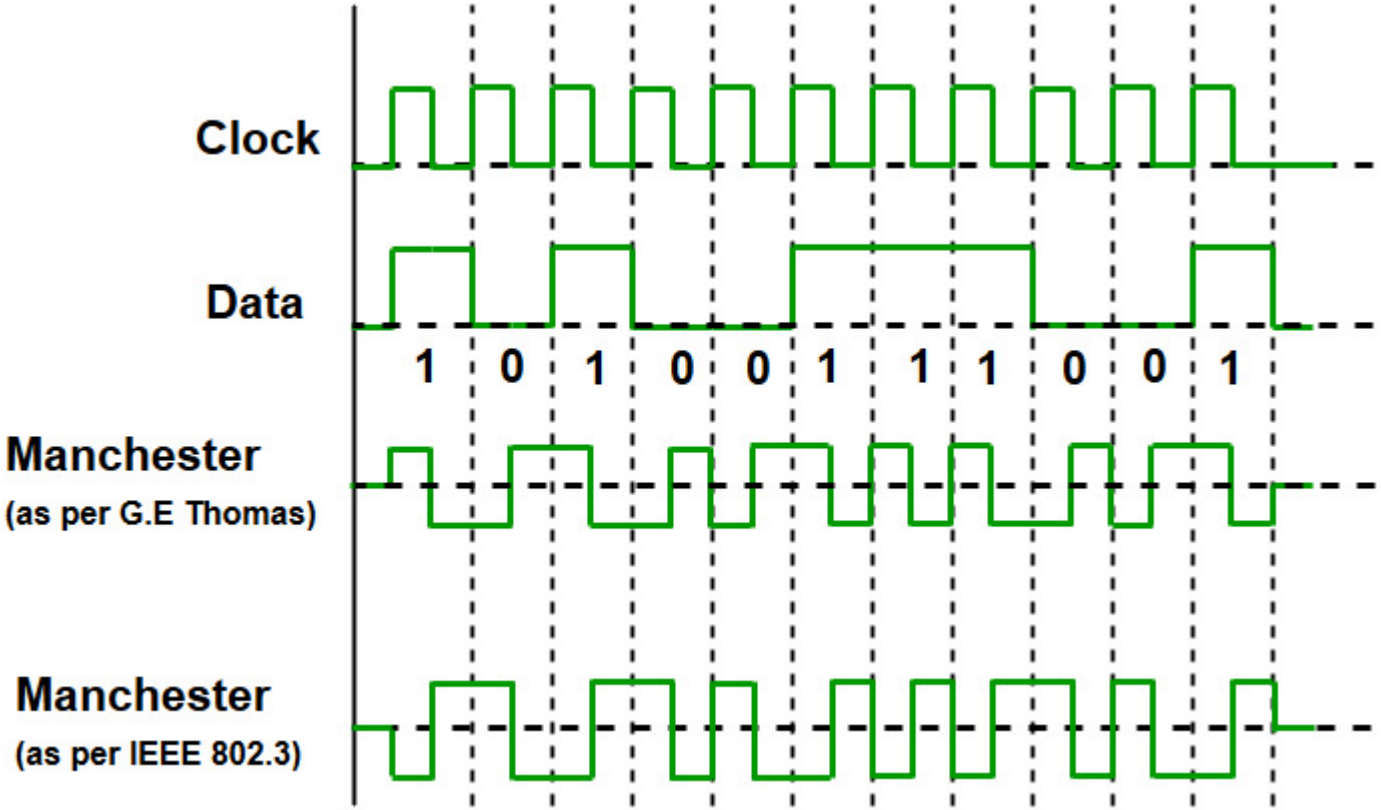


2. Data Layer

How is data transferred?

- IEEE 802.3 ➔ Ethernet

Encoding / Decoding



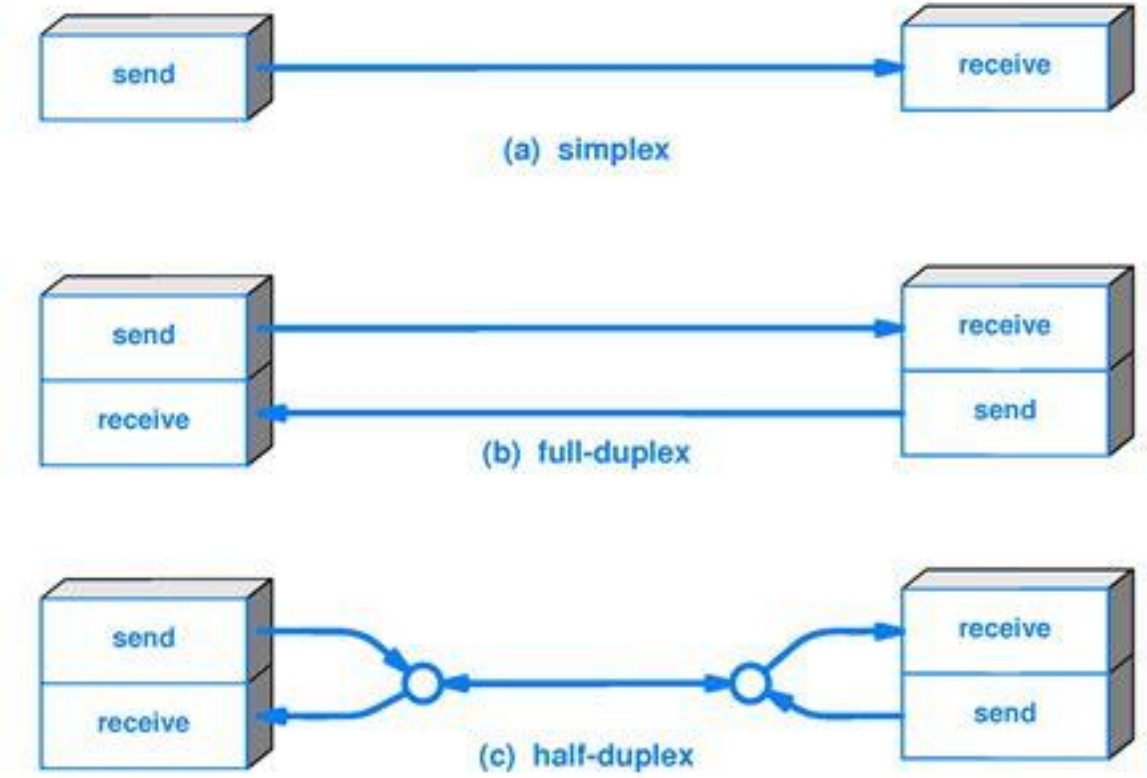
Newer Ethernet protocols have multiple levels

2. Data Layer

How is data transferred?

- IEEE 802.3 ➔ Ethernet

Transmission Modes



1. Physical Layer

How raw bits are transferred over physical media?

- For example:
  - Optical Fibers
  - Physical Cables
  - Radio Frequency
- <u>NICs and modems</u> handle data conversion and raw transmission

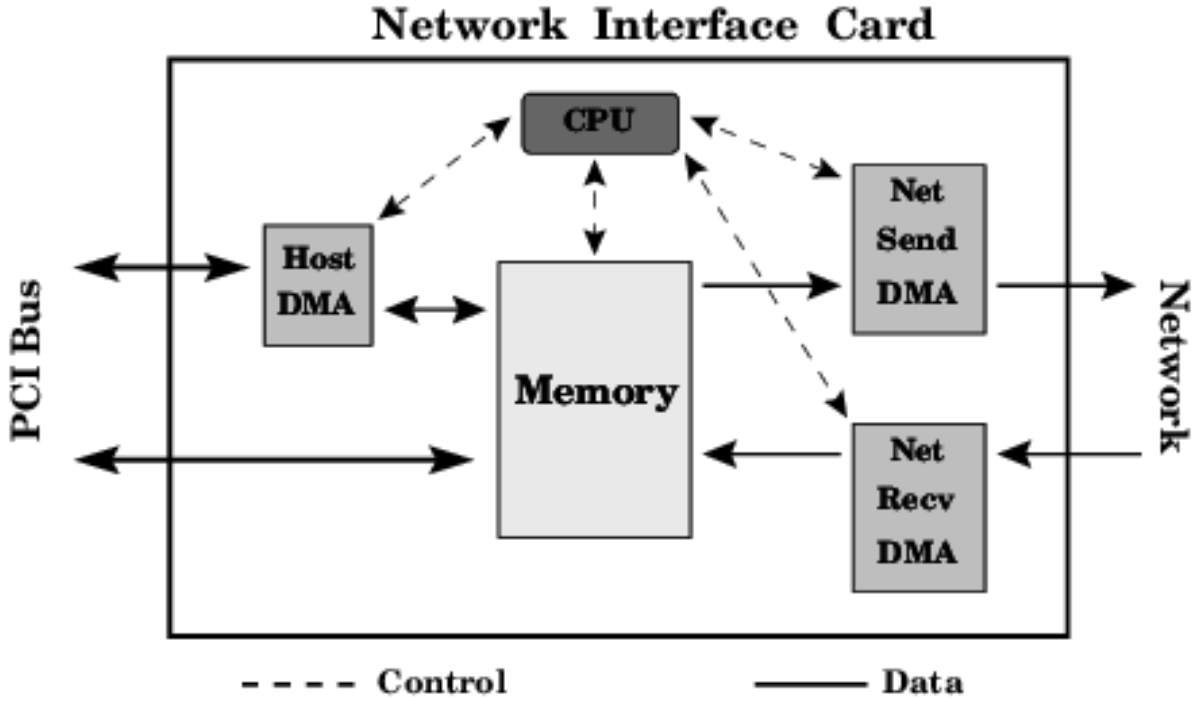
Network Devices

Network Devices

Physical Layer

Network Interface Card (NIC)

- Network interface with Driver
- Computer sends/receives data through NIC

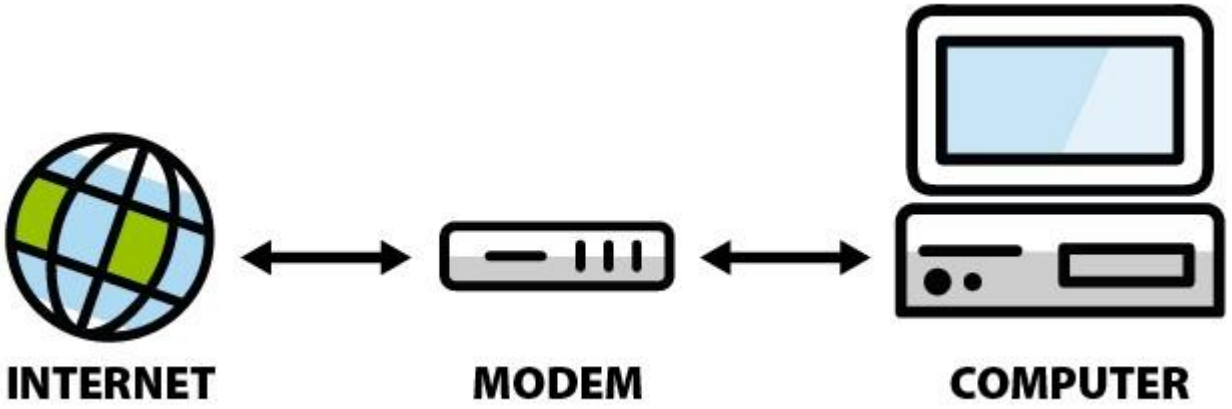


Network Devices

Physical Layer

Modem

- Modulator/Demodulator
- Converts digital data (Computer) to analogue data (Fiber/Coaxial)

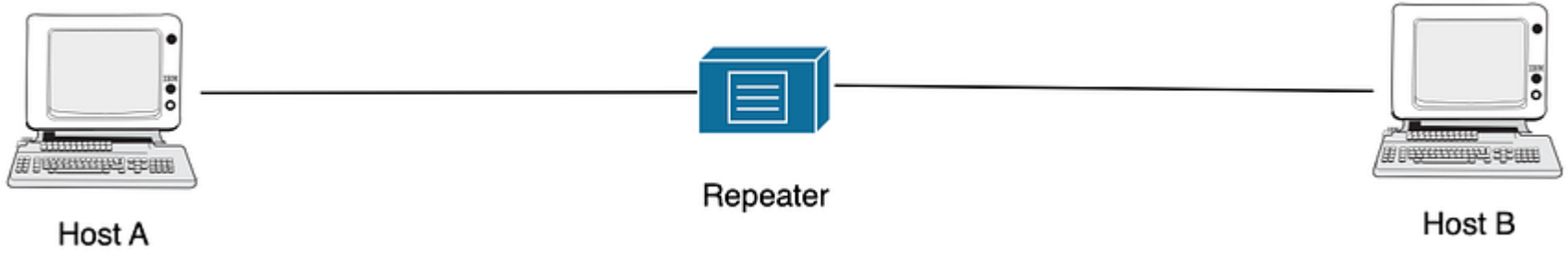


Network Devices

Physical Layer

Repeater (Extender/Booster)

- Amplifies/Regenerates the signal strength
- Within the same network

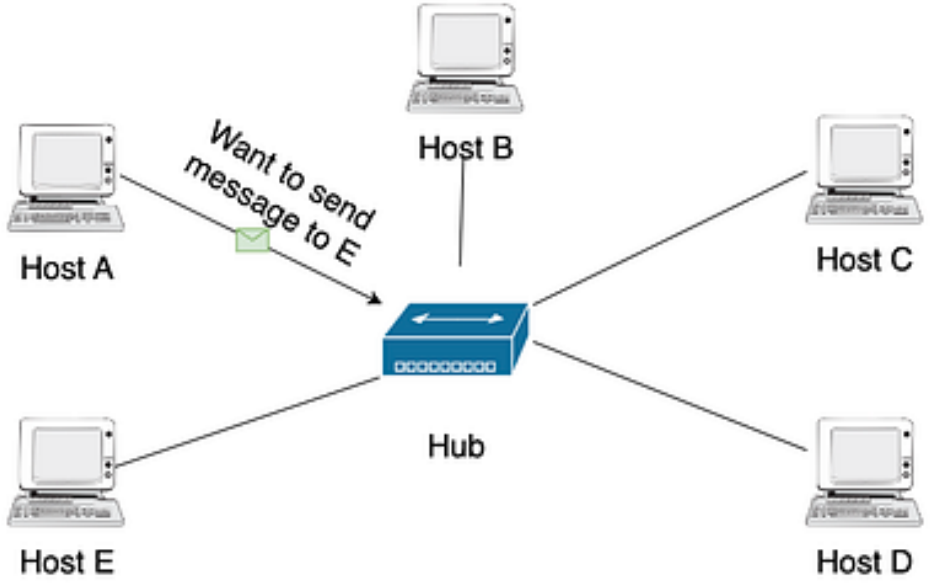


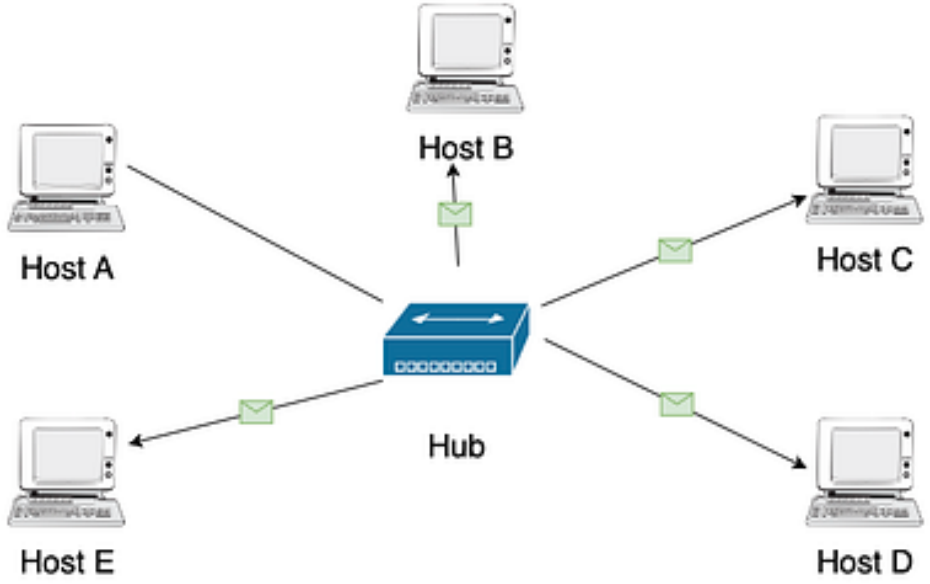
Network Devices

Physical Layer

Hub

- Multi-port Repeater
- Broadcast the data to all devices attached



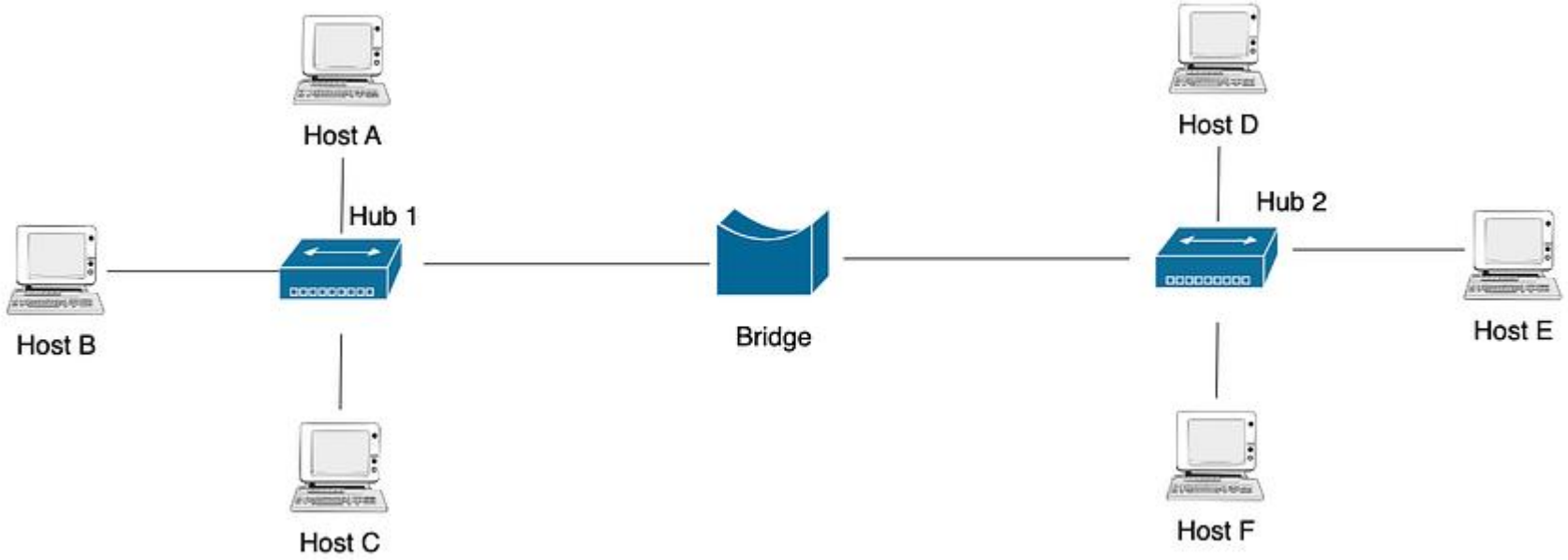


Network Devices

Data Layer

Bridge

- Two-port Connector ➔ Pass msg to the target segment
- Joins any two sub-networks or host segments

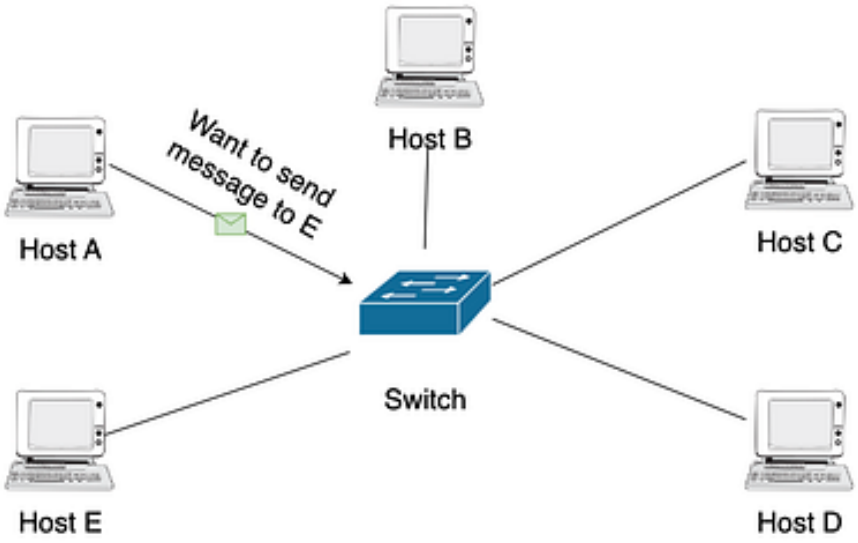


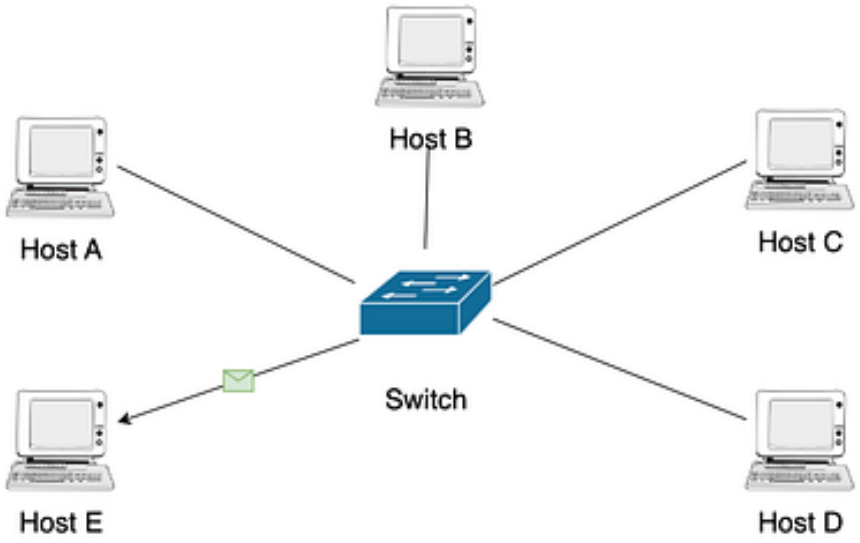
Network Devices

Data Layer

Switch

- Multi-port Connector
- Send/redirect data to the destination only



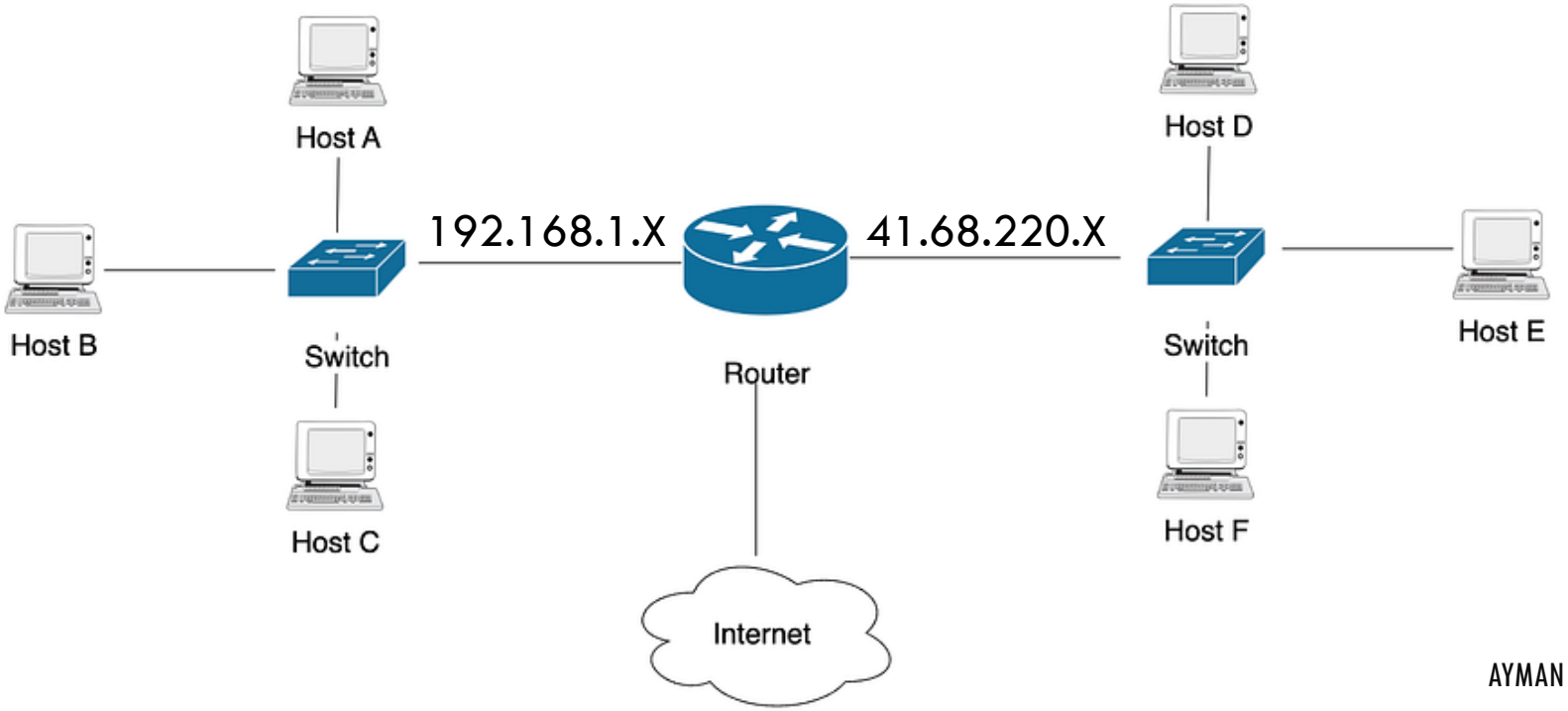


Network Devices

Network Layer

Router

- Multi-Network Connector
- Move data between networks and provide traffic control

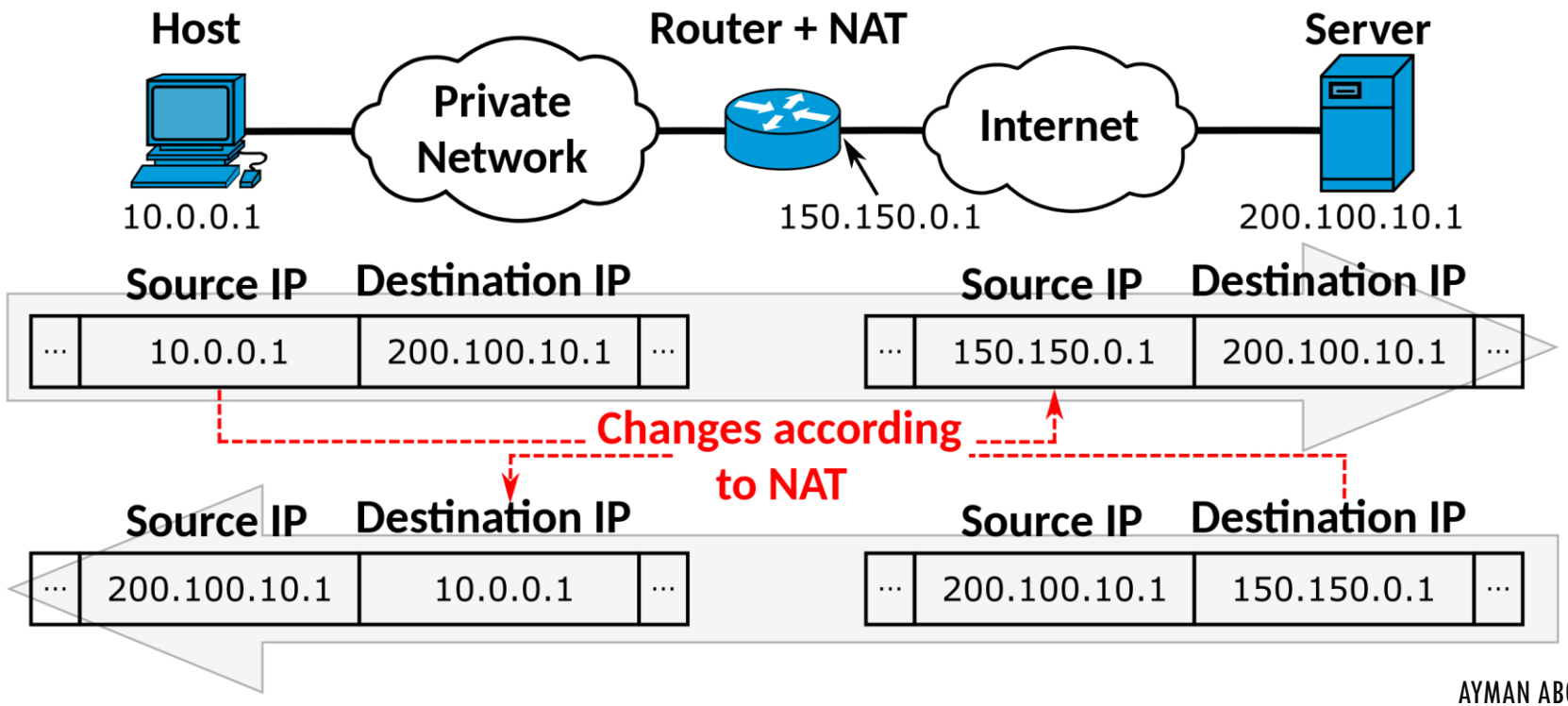


Network Devices

Network Address Translator (NAT)

Network Layer

- Translates private IPs to public IPs and vice versa

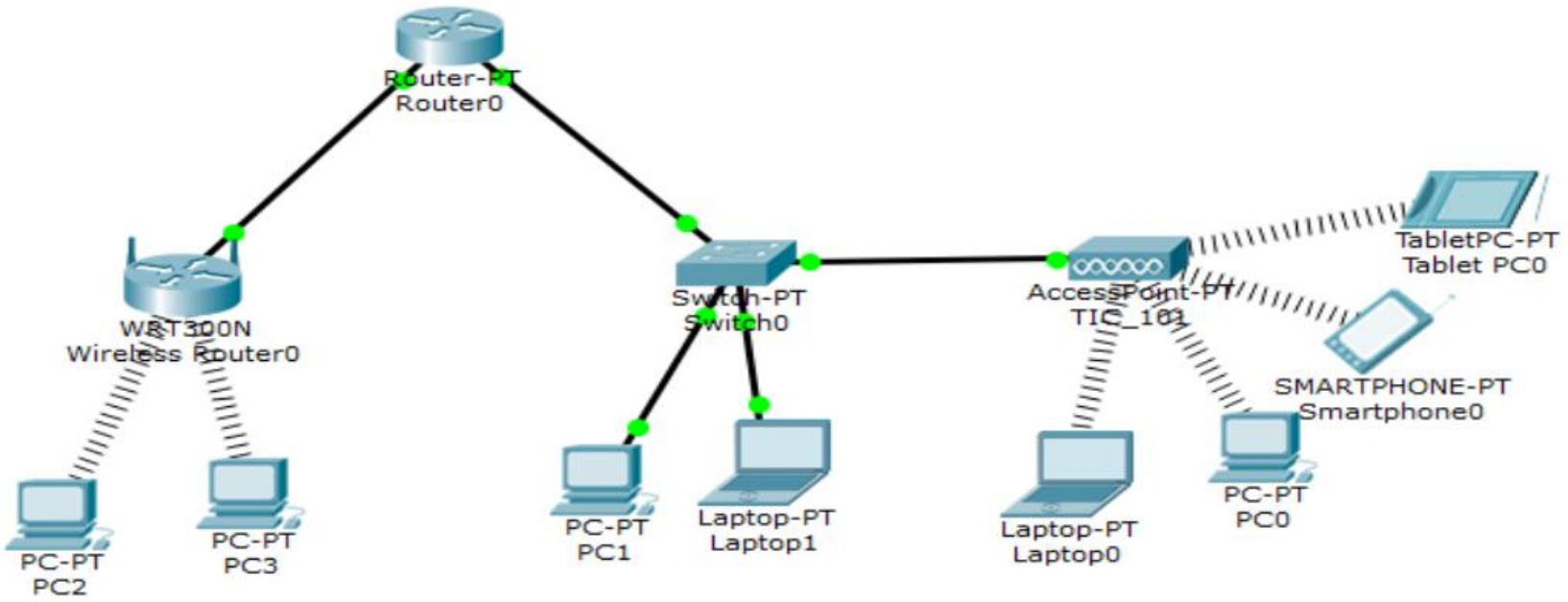


Network Devices

Network Layer

Access Point

- Router for wired/wireless devices
- Acts as a switch, DHCP server, router, and firewall

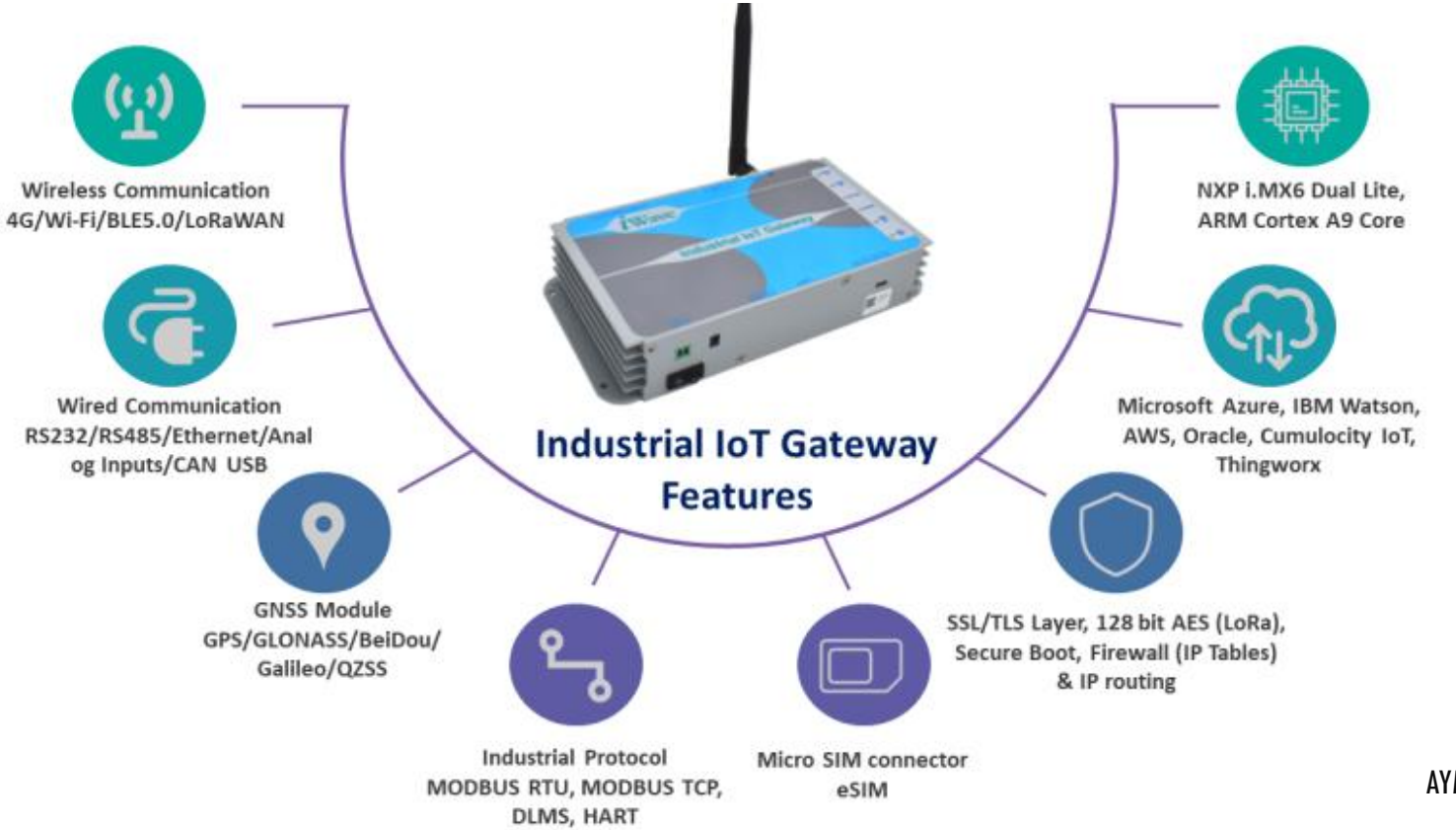


Network Devices

App Layer

Gateway

- Multi-System Connector for similar/dissimilar networks

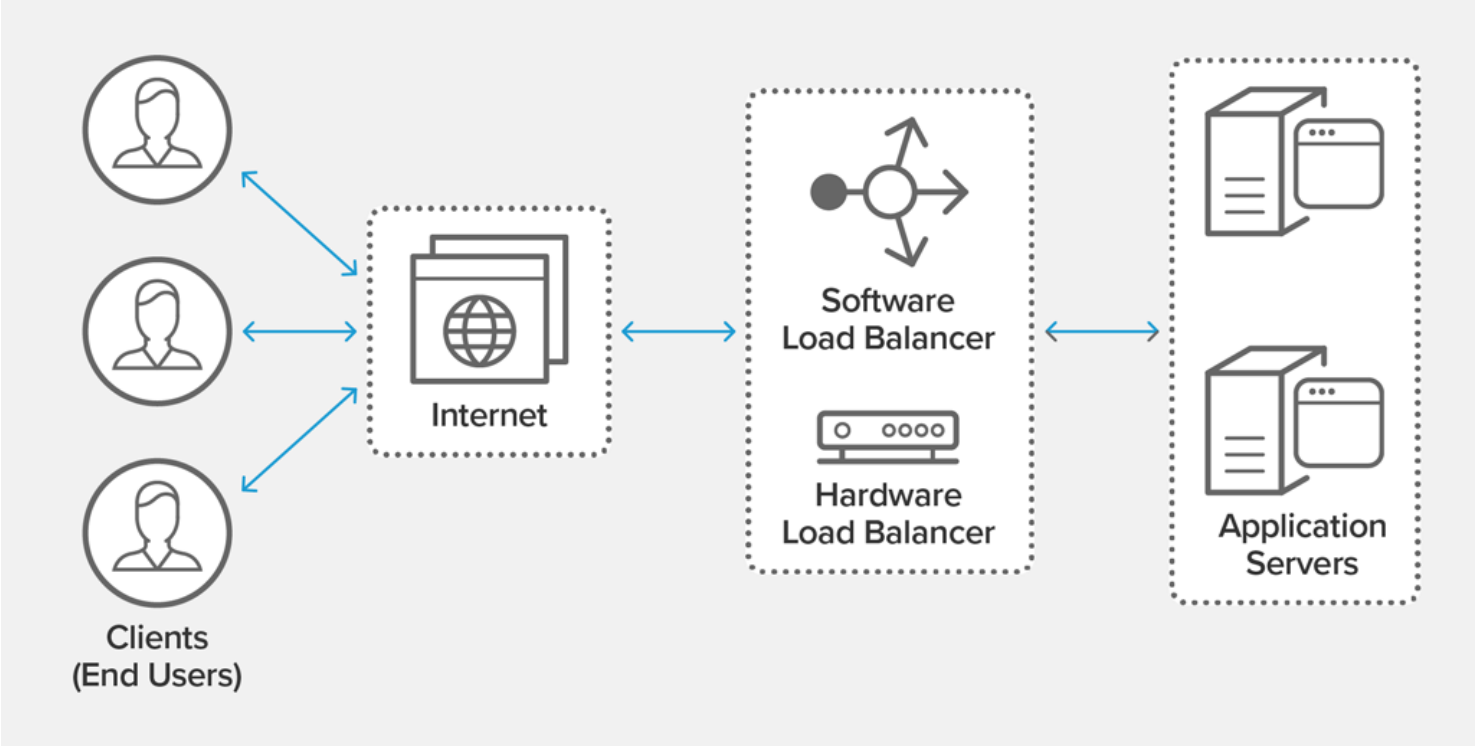


Network Devices

Load Balancer

App Layer

- Distributing incoming traffic across a group of servers

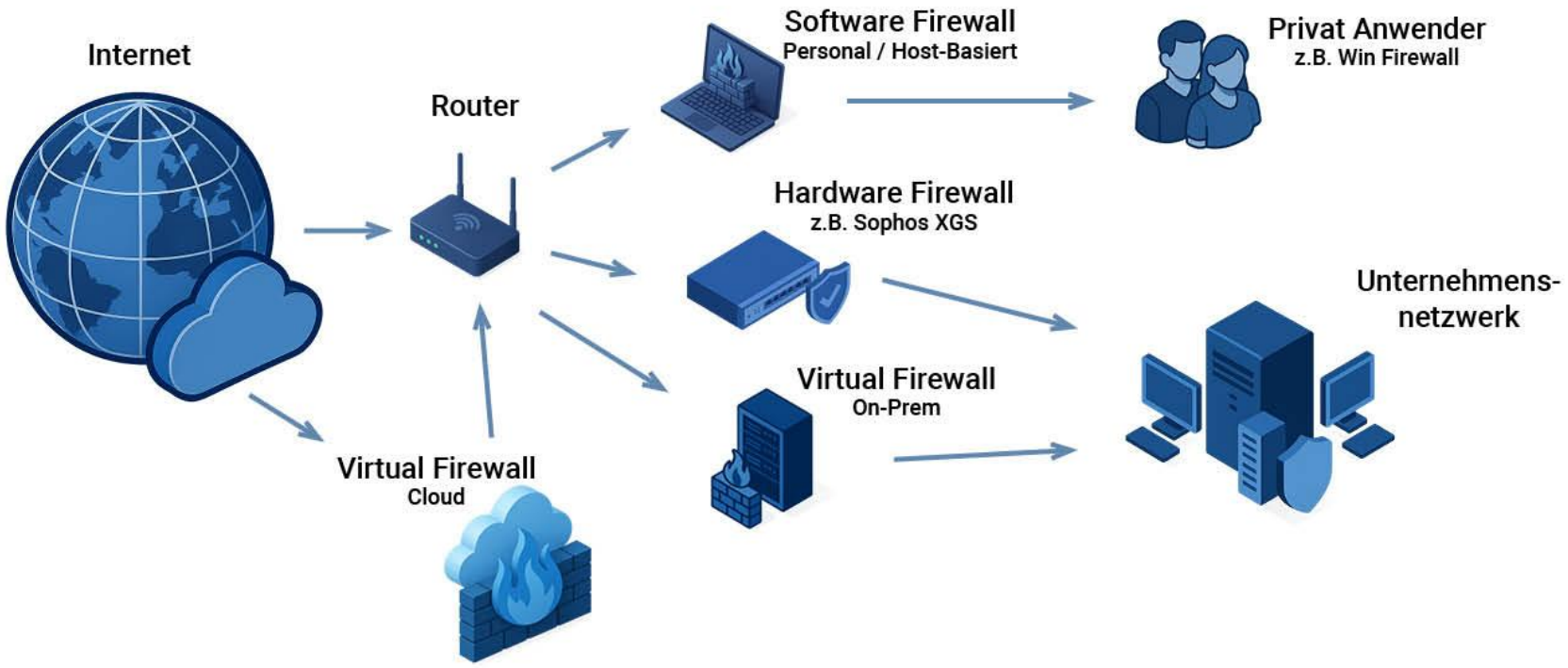


Network Devices

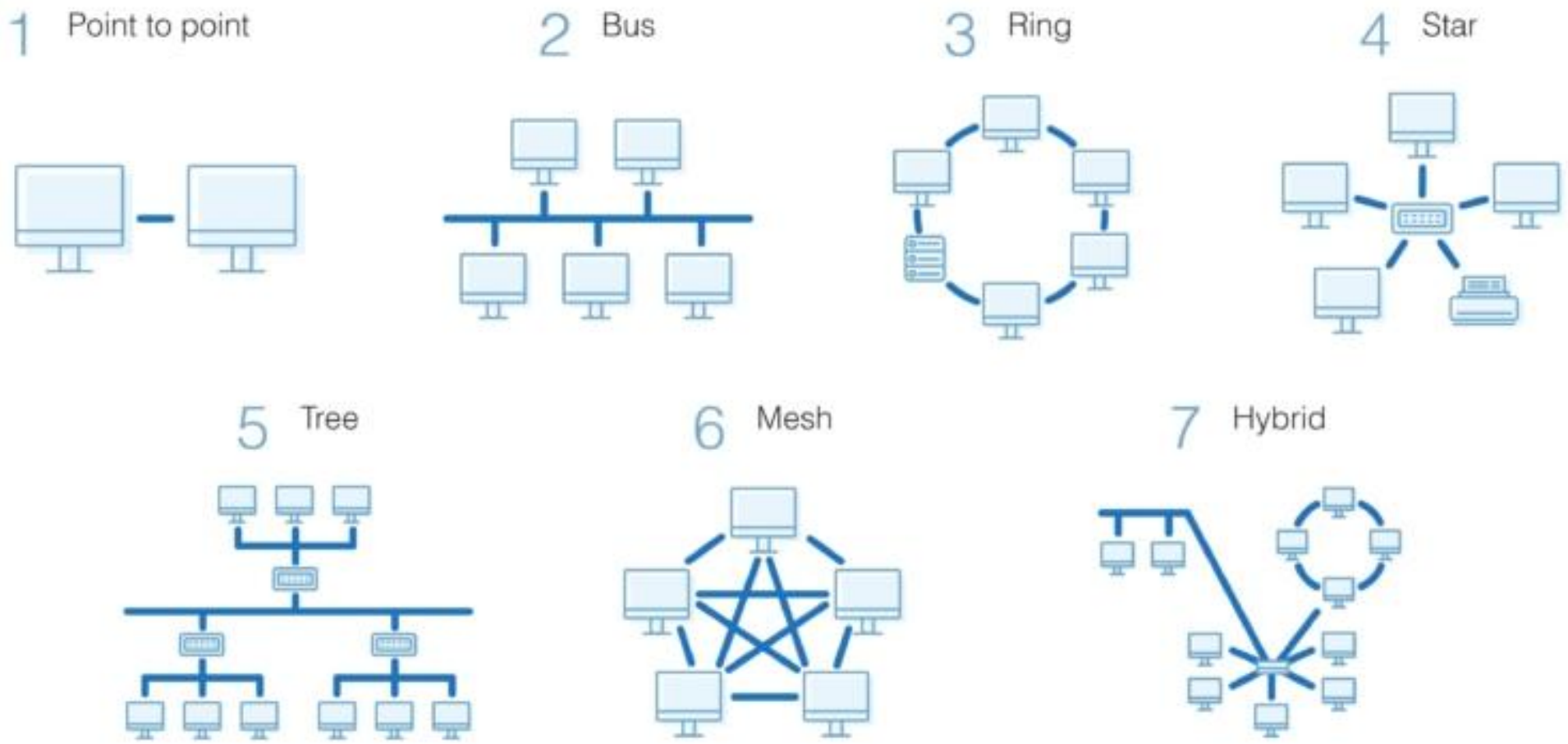
App Layer

Firewall

- Control incoming/outgoing traffic based on security rules



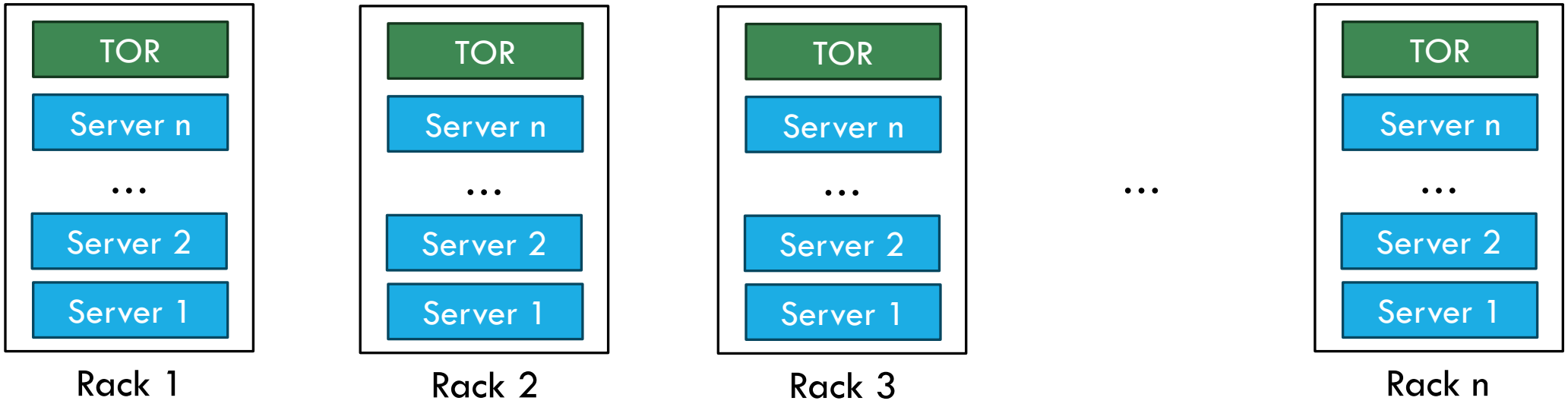
Network Topology



Cloud Network Structure

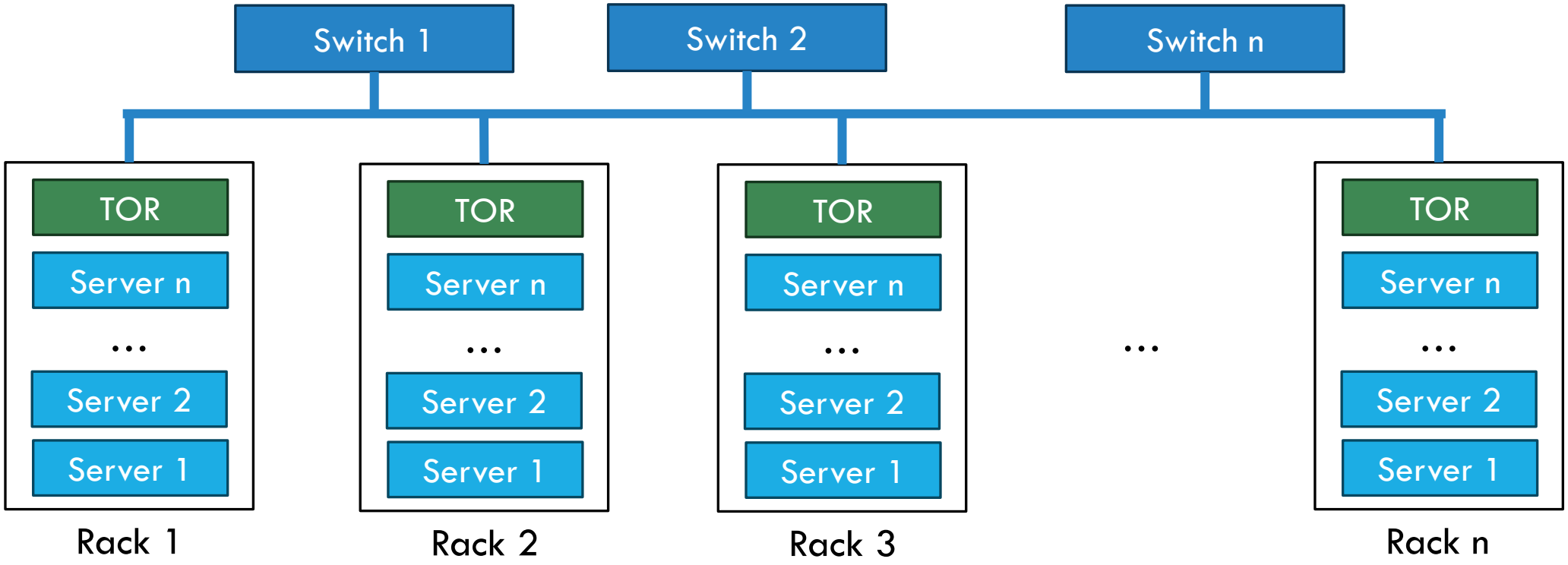
Top-of-Rack (TOR) access Switches (1 to 10Gbps)

- Connects Servers in same rack together



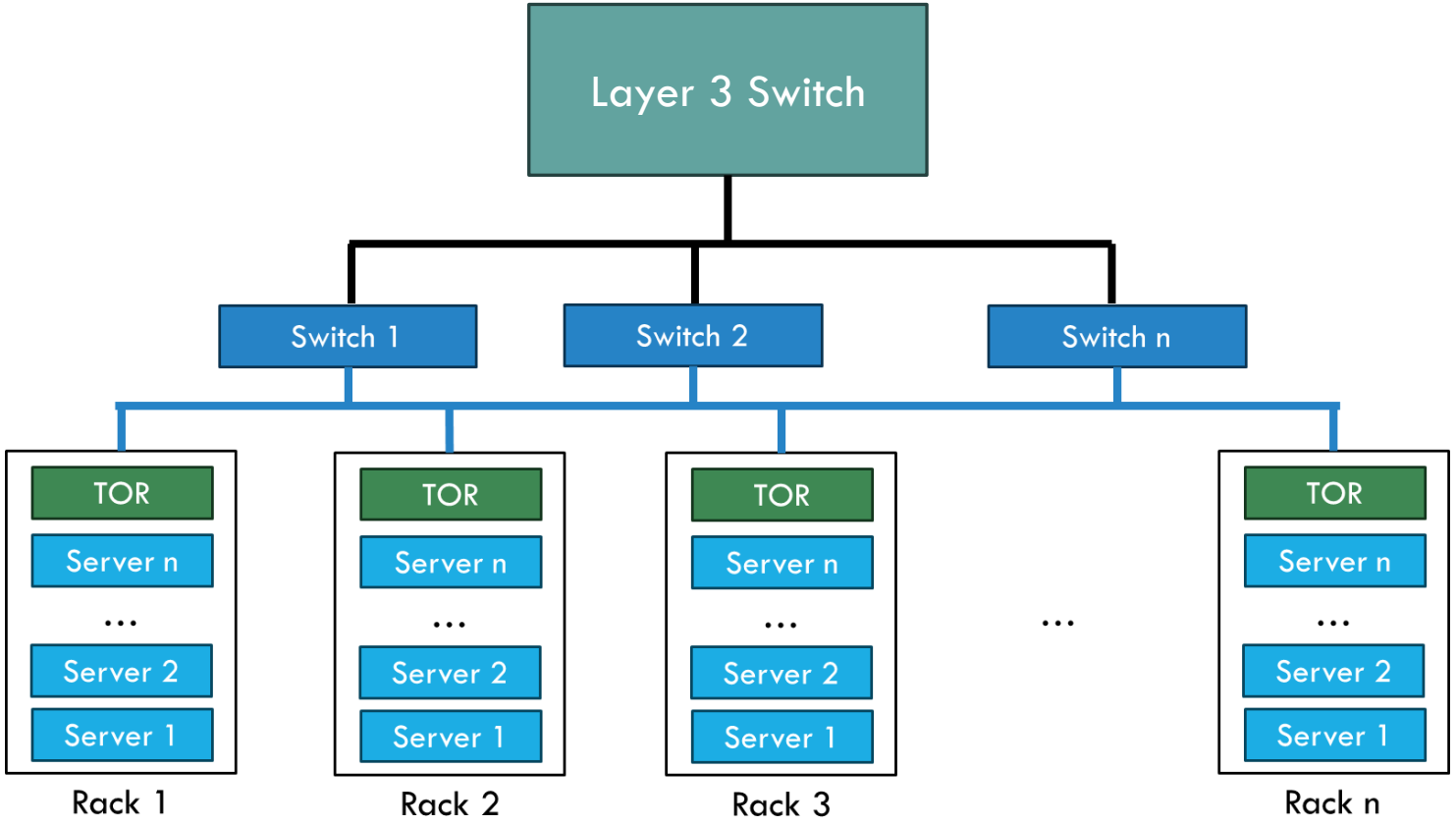
Cloud Network Structure

Core Switches (~100Gbps) ➔ Connect racks together



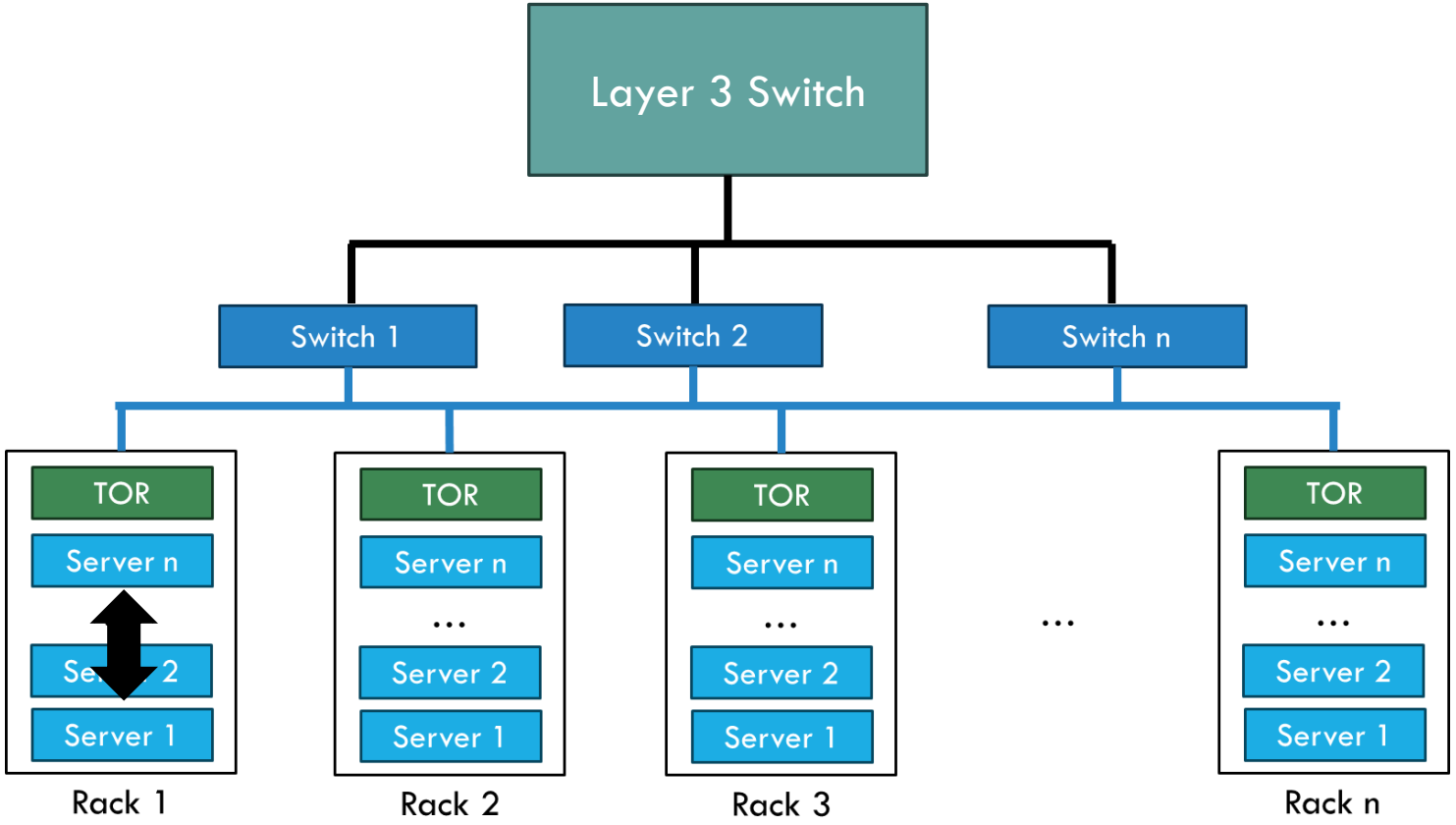
Cloud Network Structure

Layer 3 Switches/Routers ➔ Connect sectors together & internet



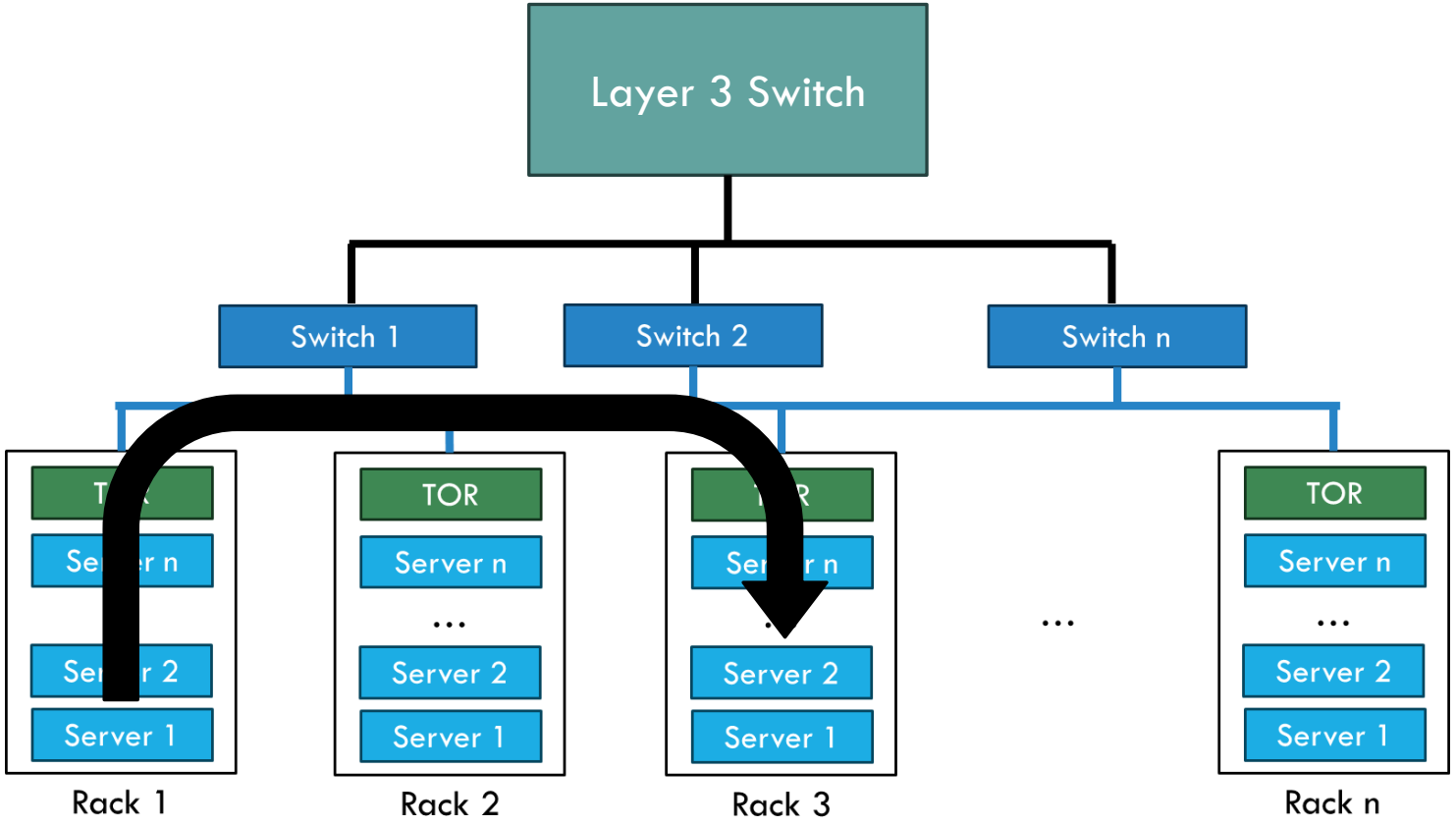
Cloud Network Structure

Traffic could be **In-Rack** (between servers in same rack)



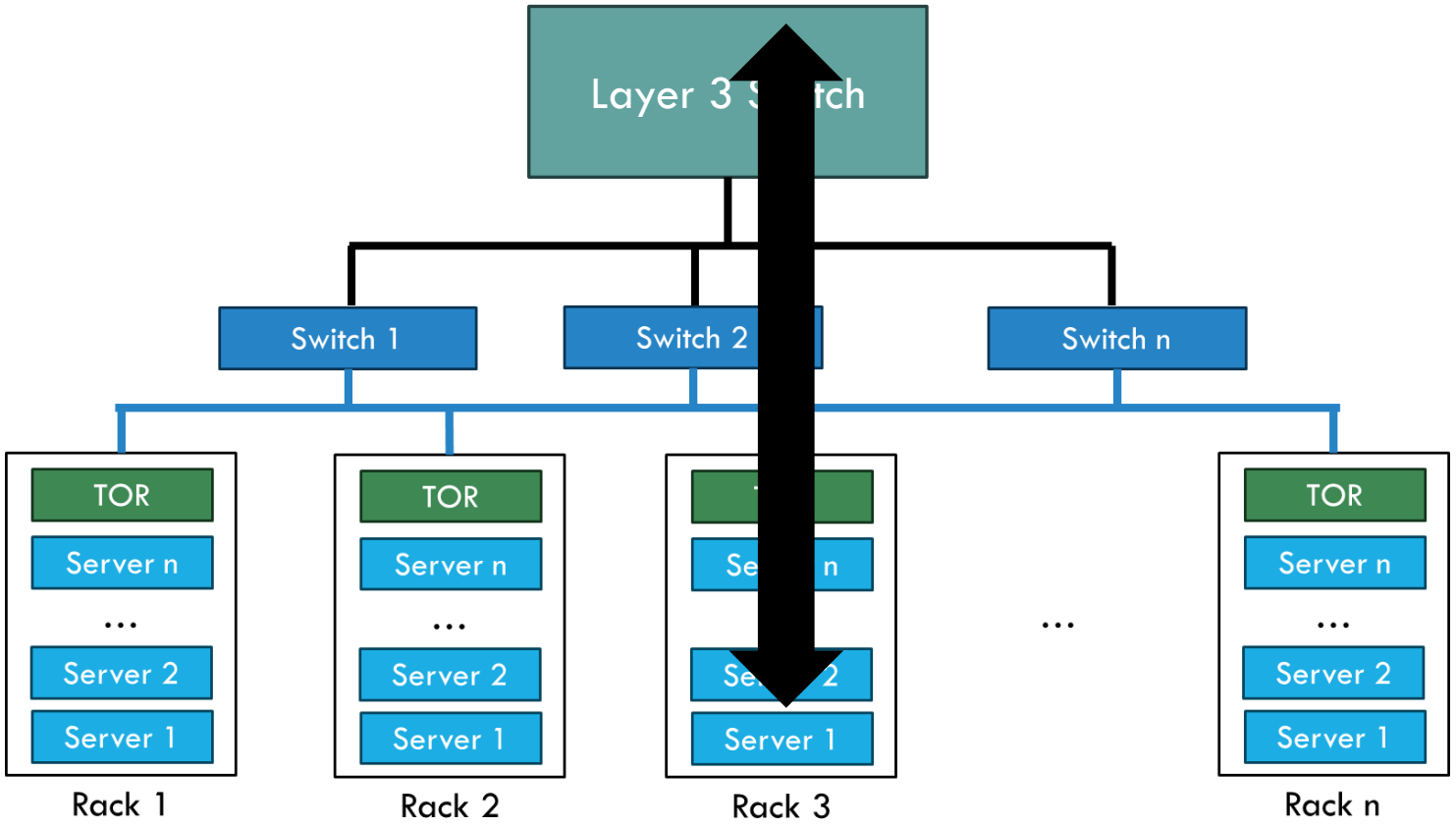
Cloud Network Structure

Traffic could be **East-West** (between servers in same sector)



Cloud Network Structure

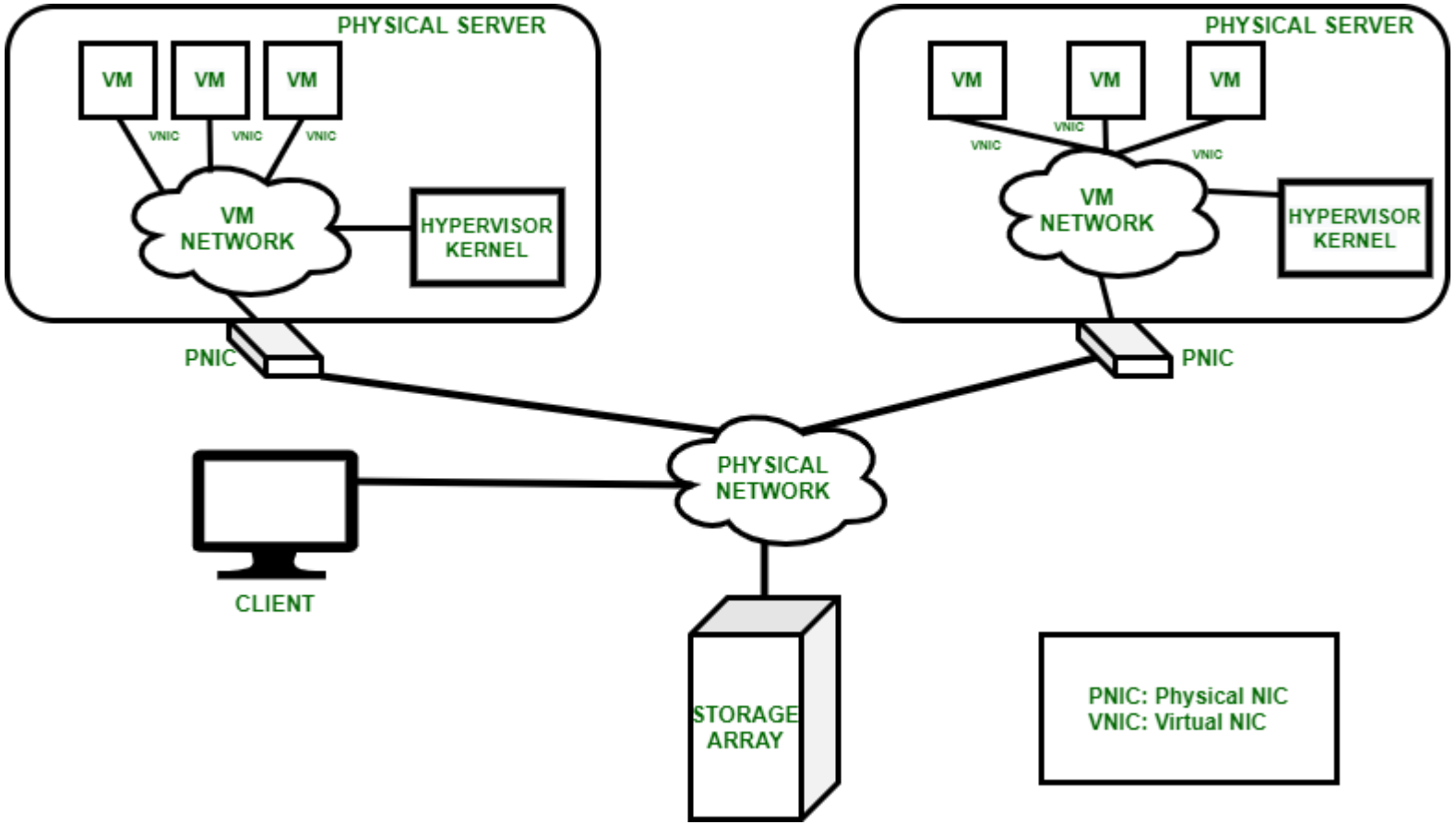
Traffic could be **North-South** (between sectors / internet)



Network Virtualization

Inside Cloud Data Centers

Virtual entities (VMs/Containers) running inside physical servers, <u>communicating</u> with each <u>other</u> & with the <u>internet</u>



Inside Cloud Data Centers

1. Create logical “Virtual” networks between VMs
2. Virtual networks are isolated
3. Virtual networks could share same underlying physical resources (i.e.: links, routers, switches)

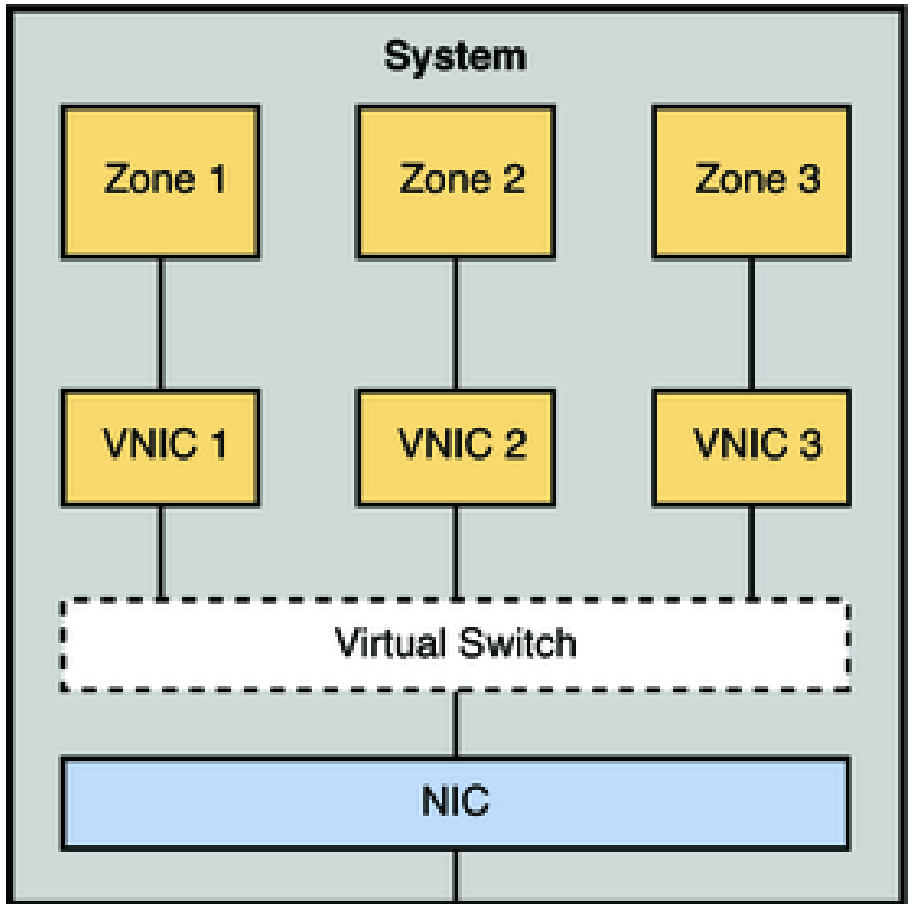
Network Hypervisor

Abstraction layer

- Represents nodes and links in a virtual network
- Control resources (network devices i.e.: Routers, Switches)
- Control bandwidth
- Control virtual network capacity
- Isolate & secure virtual networks

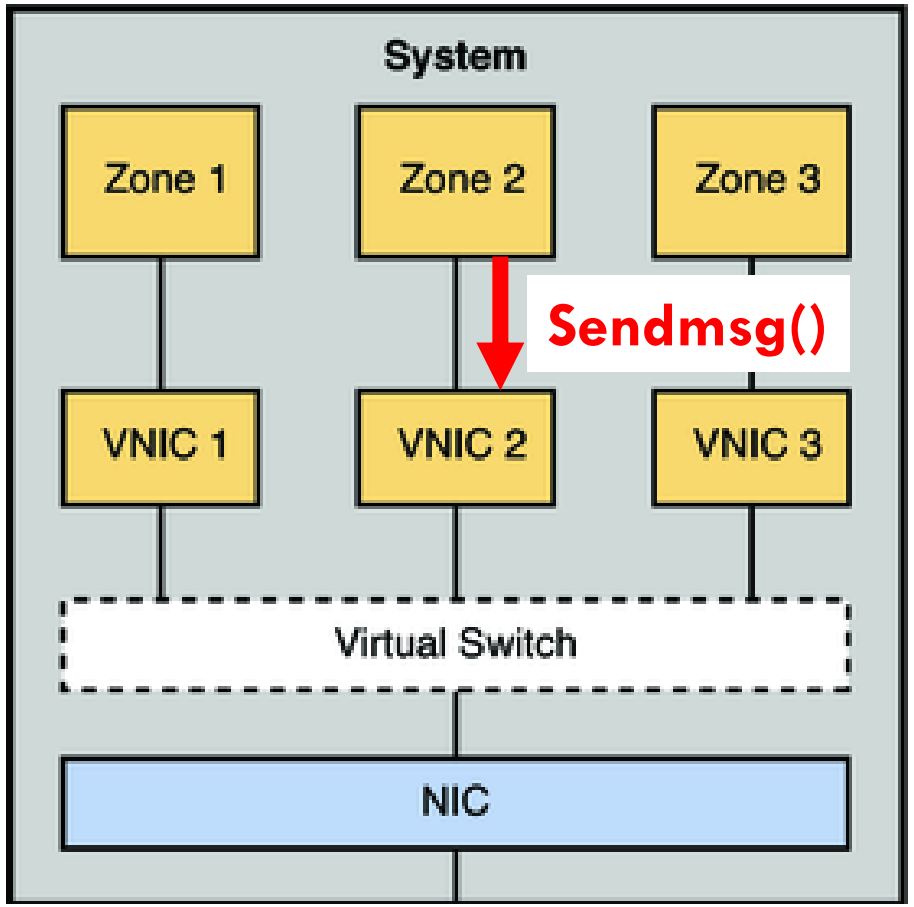
Types of Network Virtualization

1. Internal Virtualization
   - How VMs on the same server communicate?
     - **VMM Hypervisor** creates virtual NICs and virtual switches
     - Hypervisor kernel provides connectivity
     - Virtual network resides inside the physical server



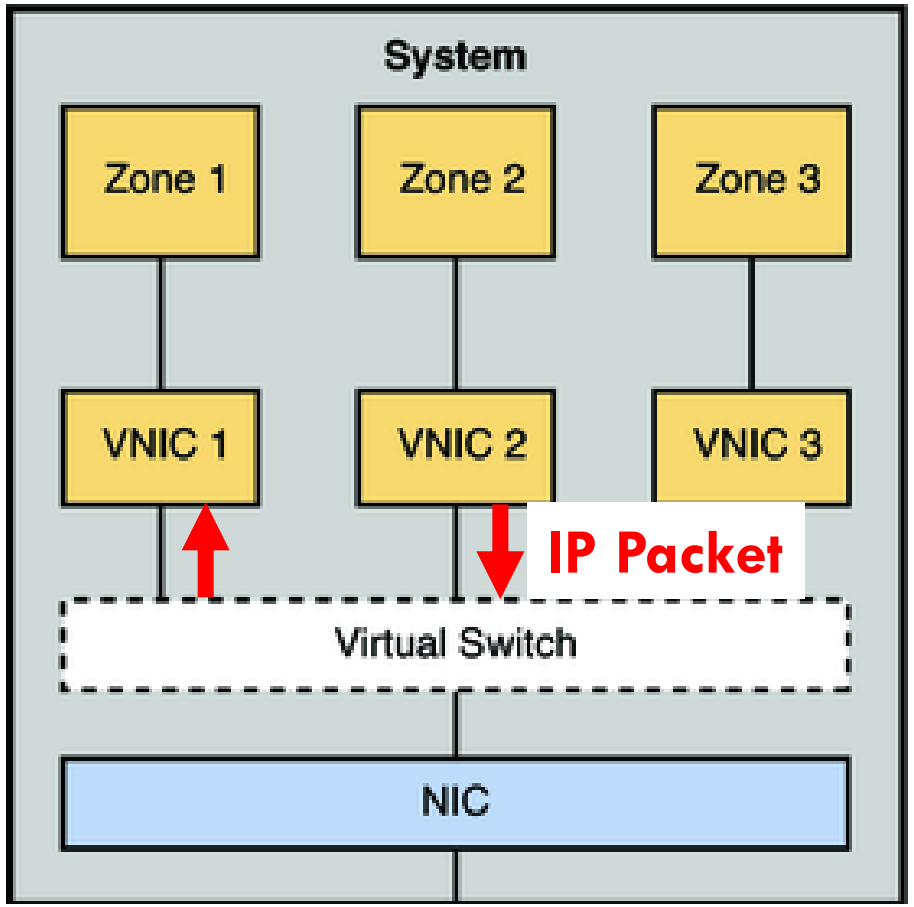
Types of Network Virtualization

1. Internal Virtualization
   1. Emulate System Calls
      - When VM triggers a network system call (i.e.: bind, socket, sendmsg, recvmsg, …)
      - Hypervisor **traps** the system call & **emulates** operation using Virtual NICs
      - Virtual NIC defines the VM connectivity and bandwidth



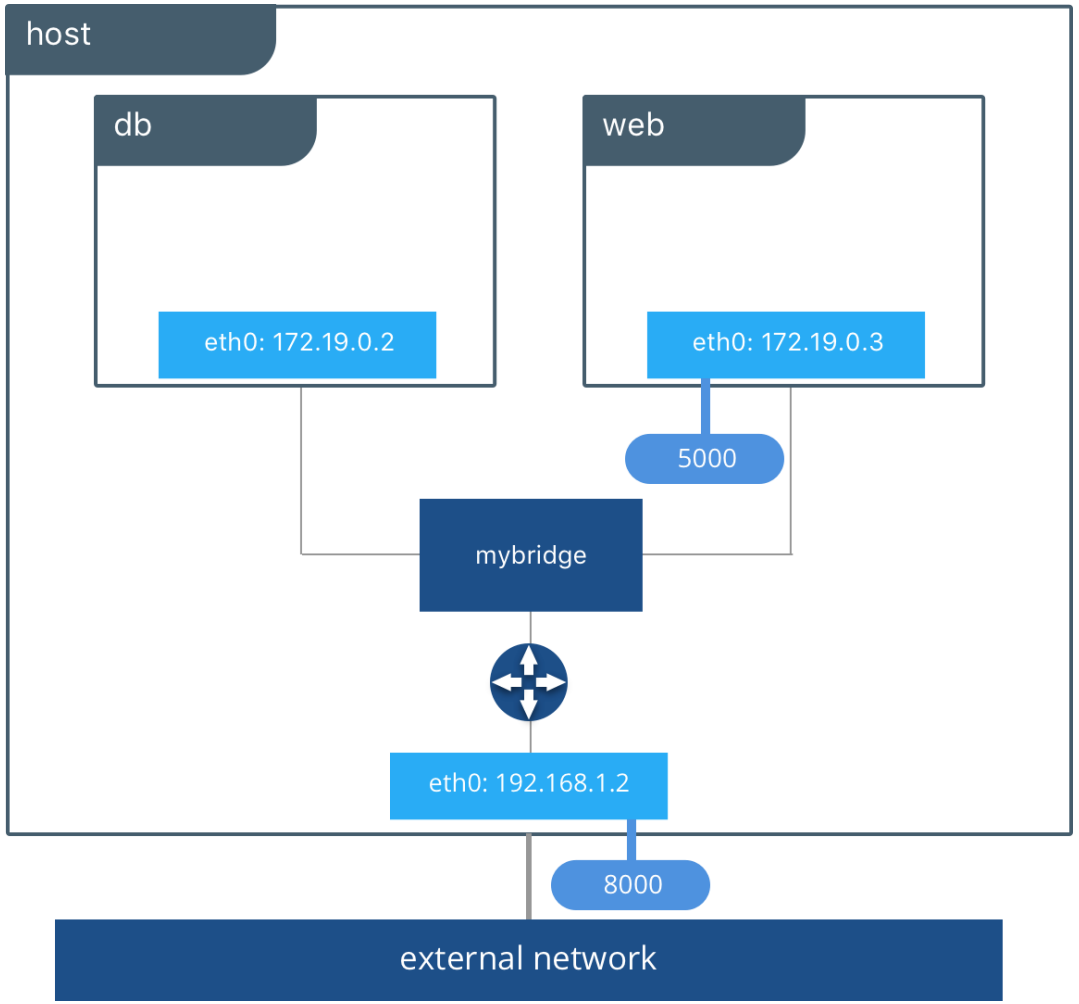
Types of Network Virtualization

1. Internal Virtualization
2. Emulate Network Function
   - Network requests are forwarded to Hypervisor's virtual switch
   - Virtual switch **routes** network messages to target Virtual NIC



Types of Network Virtualization

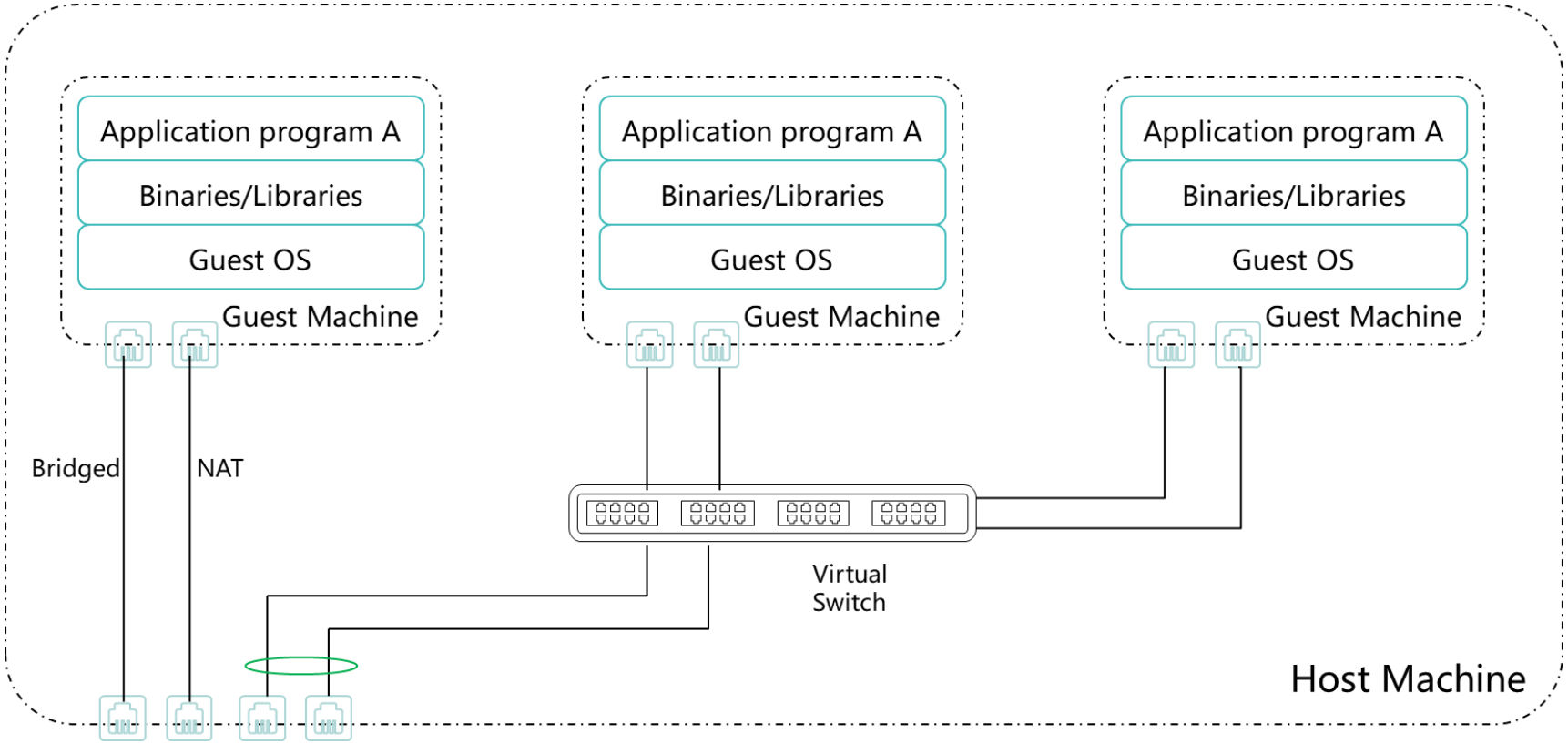
1. Internal Virtualization
   - For Example:
     1. VMs/Containers created for web app (backend, frontend)
     2. Docker/Hypervisor creates a virtual network with subnet 172.19.0.0/16 for VMs
     3. Docker/Hypervisor creates a virtual bridge to connect to the virtual network to external LAN



https://docs.docker.com/engine/network/

Types of Network Virtualization

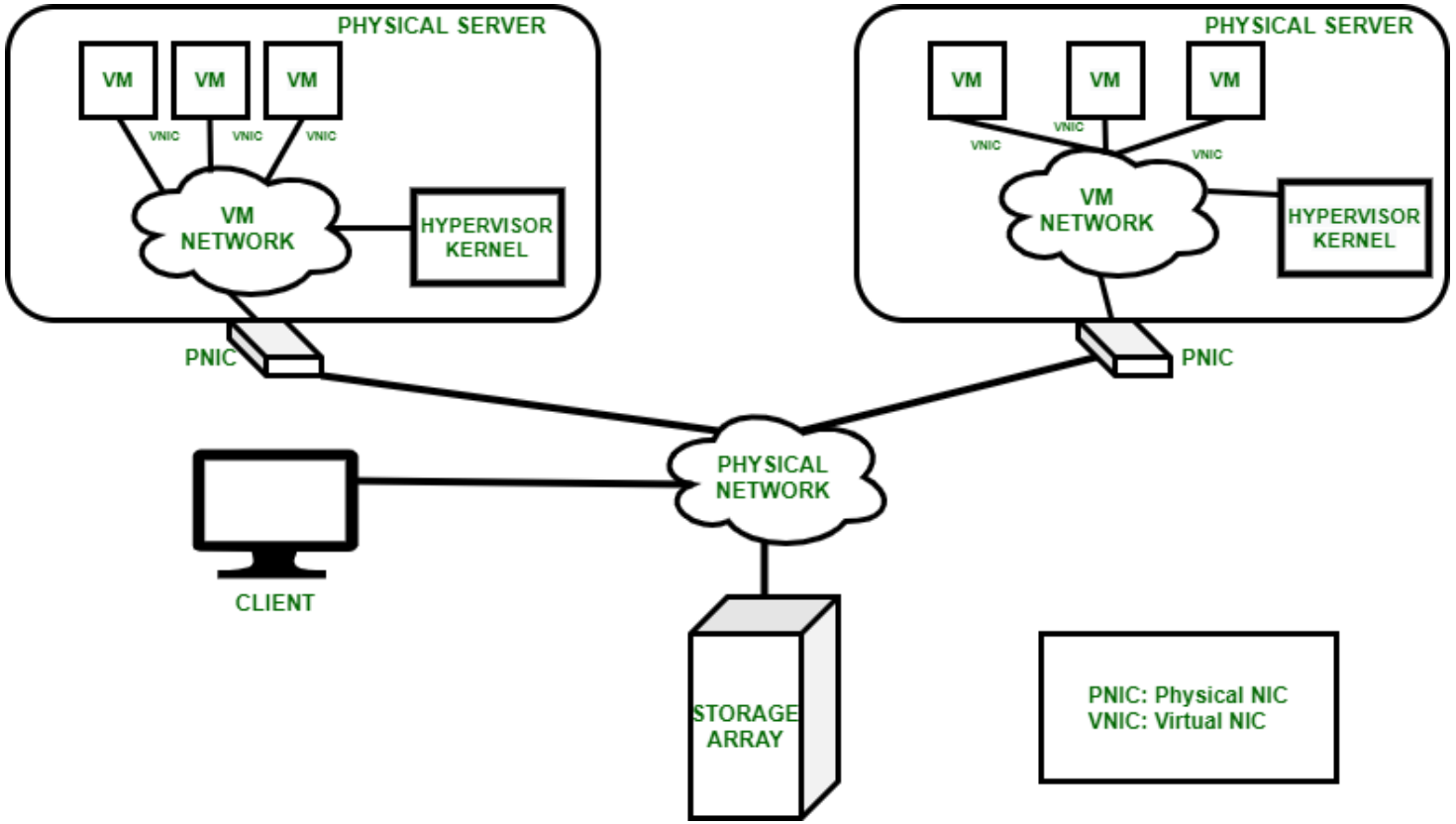
1. Internal Virtualization



Types of Network Virtualization

2. External Virtualization

- How VMs on different servers communicate?
  - Hypervisor holds virtual iptables with physical servers hosting the target VMs
  - Network devices could be aware of virtualization

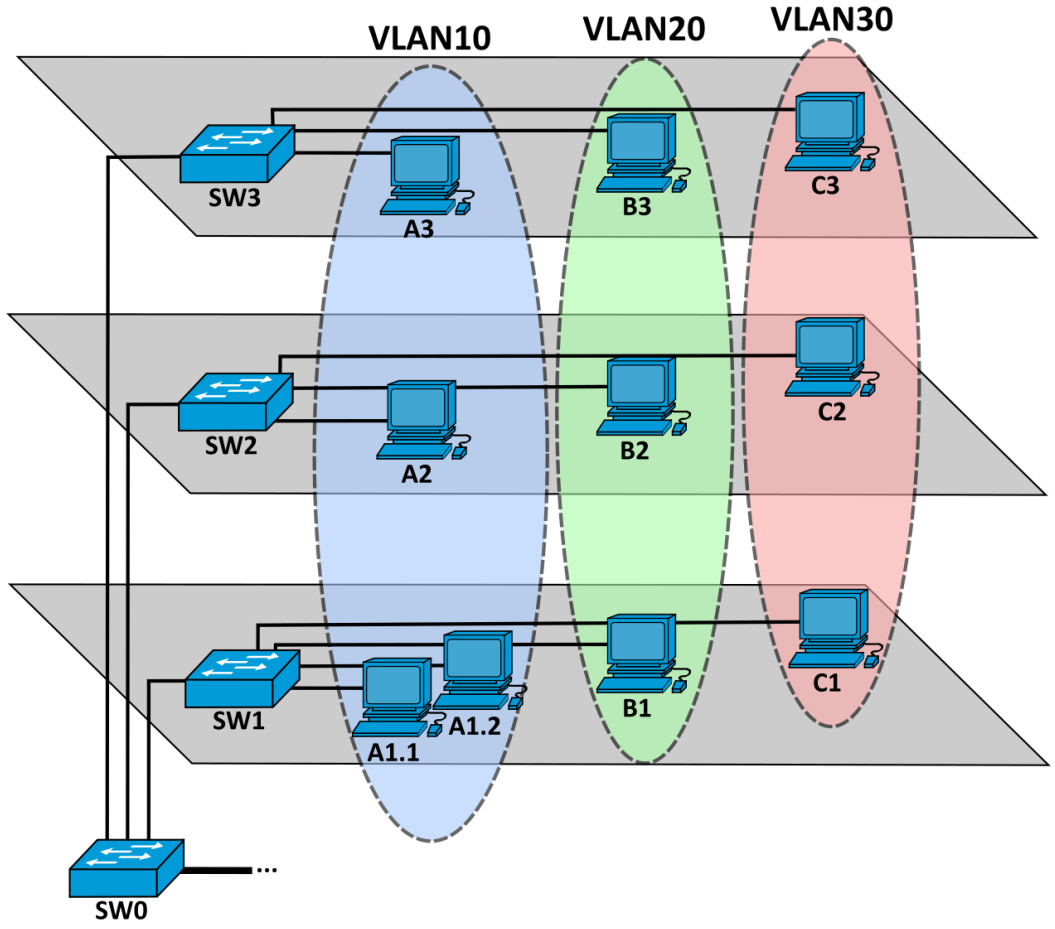


External Network Virtualization

Virtual LAN (VLAN)

Group devices into separate groups

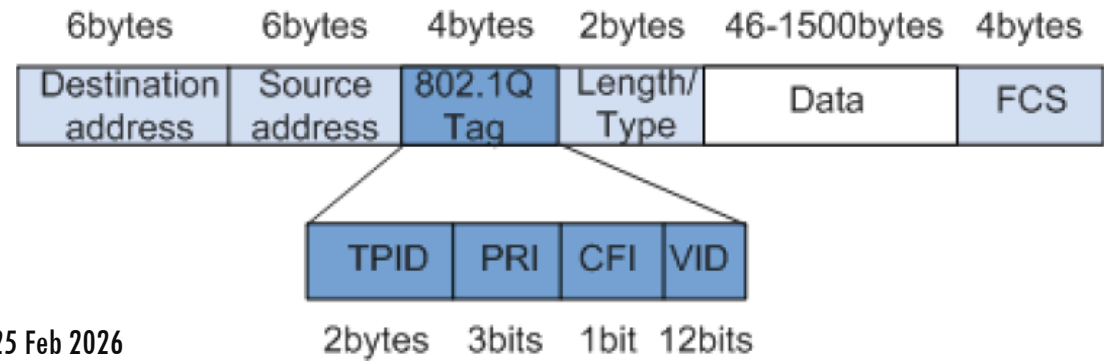
- System administrator configure the network into multiple sub-groups isolated from each other

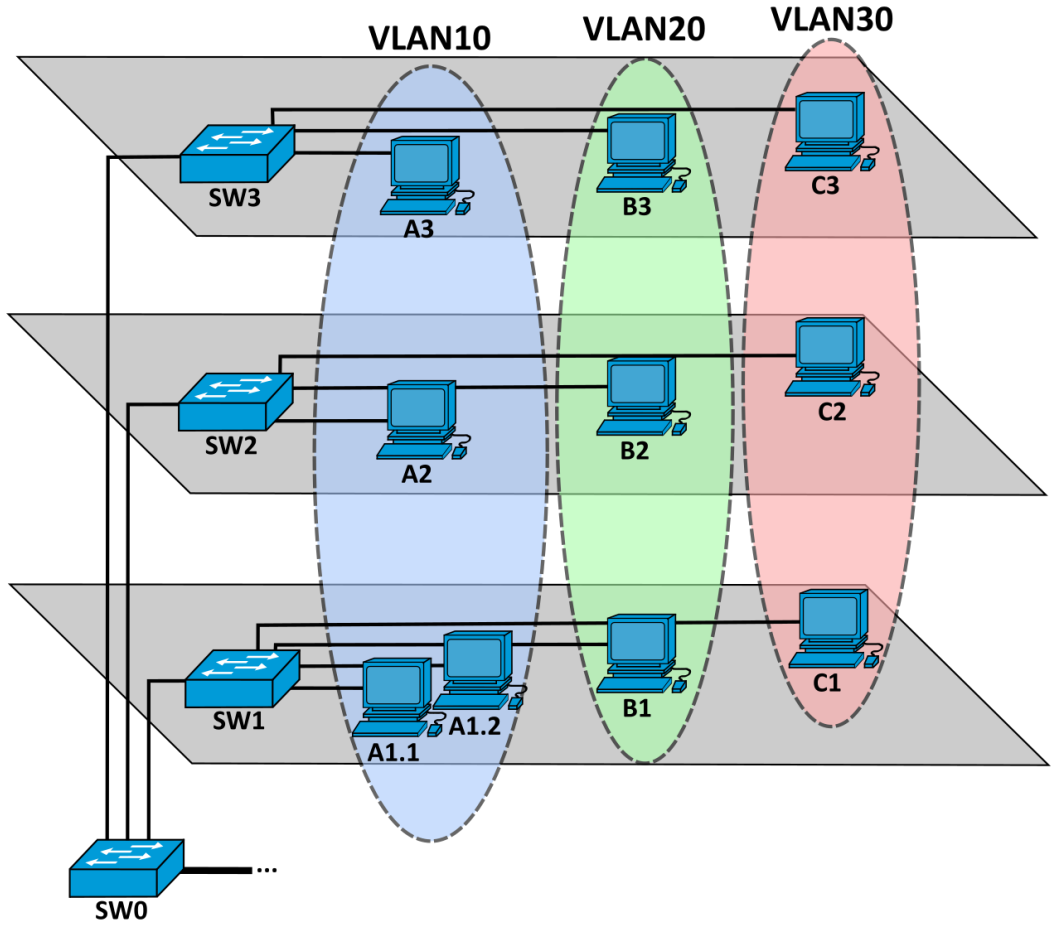


Virtual LAN (VLAN)

Group devices into separate groups

- System administrator configure the network into multiple sub-groups isolated from each other
- Layer 2 Switches route messages between devices according to VLAN group
  - IEEE 802.1Q Ethernet Standard

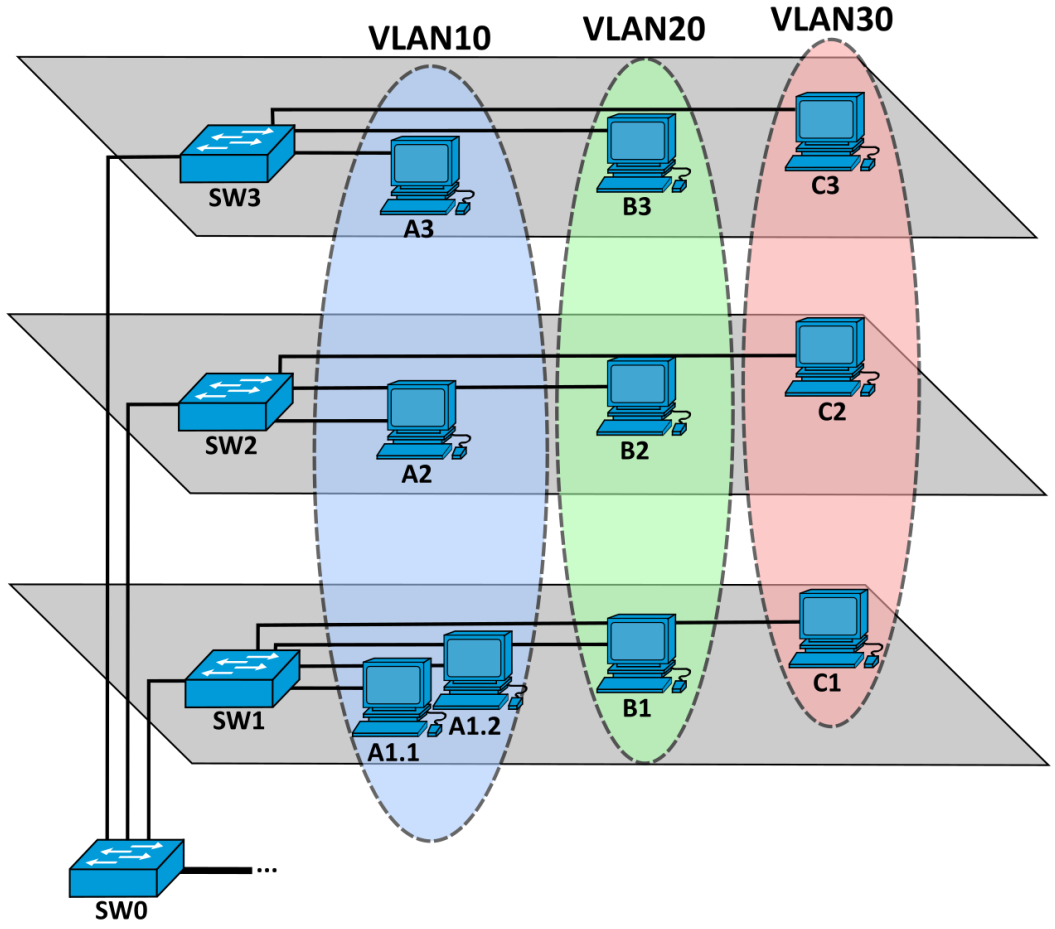




Virtual LAN (VLAN)

Group devices into separate groups

- Switches are **virtualization-aware**
- Devices in the same VLAN forms a group and communicate directly
- Devices from different VLANs require inter-VLAN routing to communicate



Virtual LAN (VLAN)

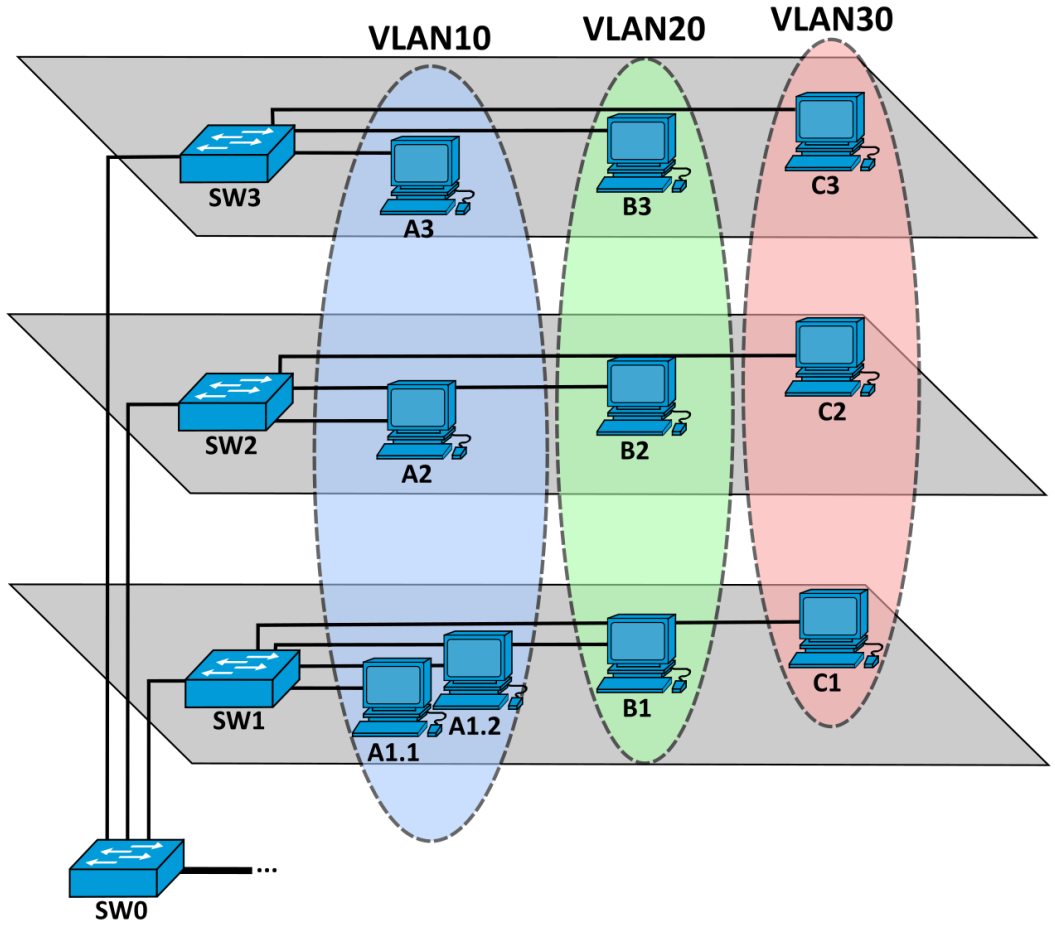
Group devices into separate groups

Pros

- Security: Network traffic isolation
- Scalability: Large network segmentation

Cons

- Complexity: Requires careful configuration
- Interoperability: Compatibility with legacy / non-aware network devices

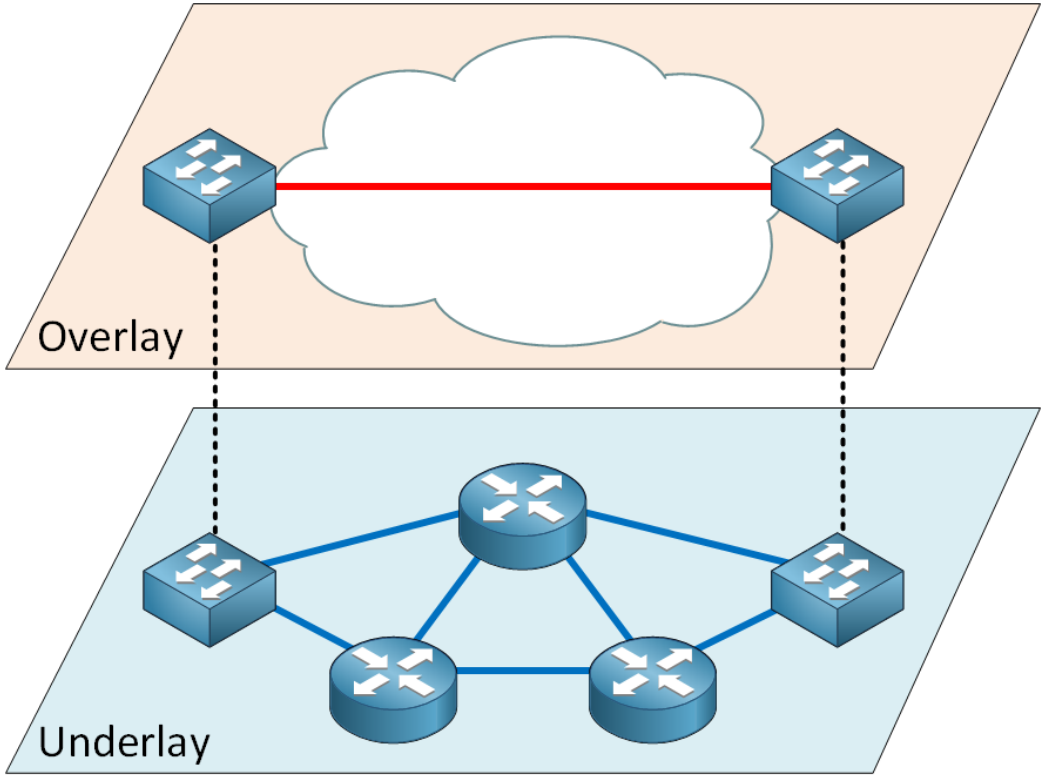


Virtual Extensible LAN (VXLAN)

Addressing Complexity Problem

- VLAN ID in IEEE 802.1Q is defined by 12 bits (4096 segments)
  - Not enough for Cloud data centers
  - Each VM has VLAN address
  - Too much manual configuration
  - Scalability ↓

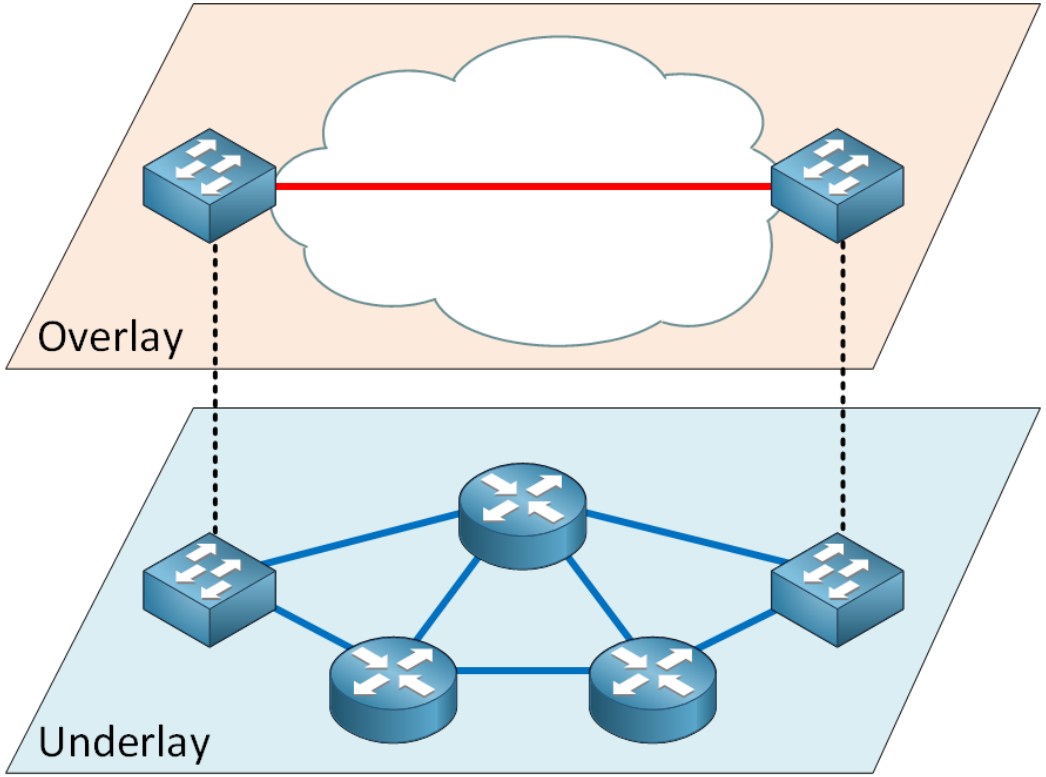
Solution ➔ VXLAN



Virtual Extensible LAN (VXLAN)

Virtualize VLAN using Layer-3 Switches/Routers

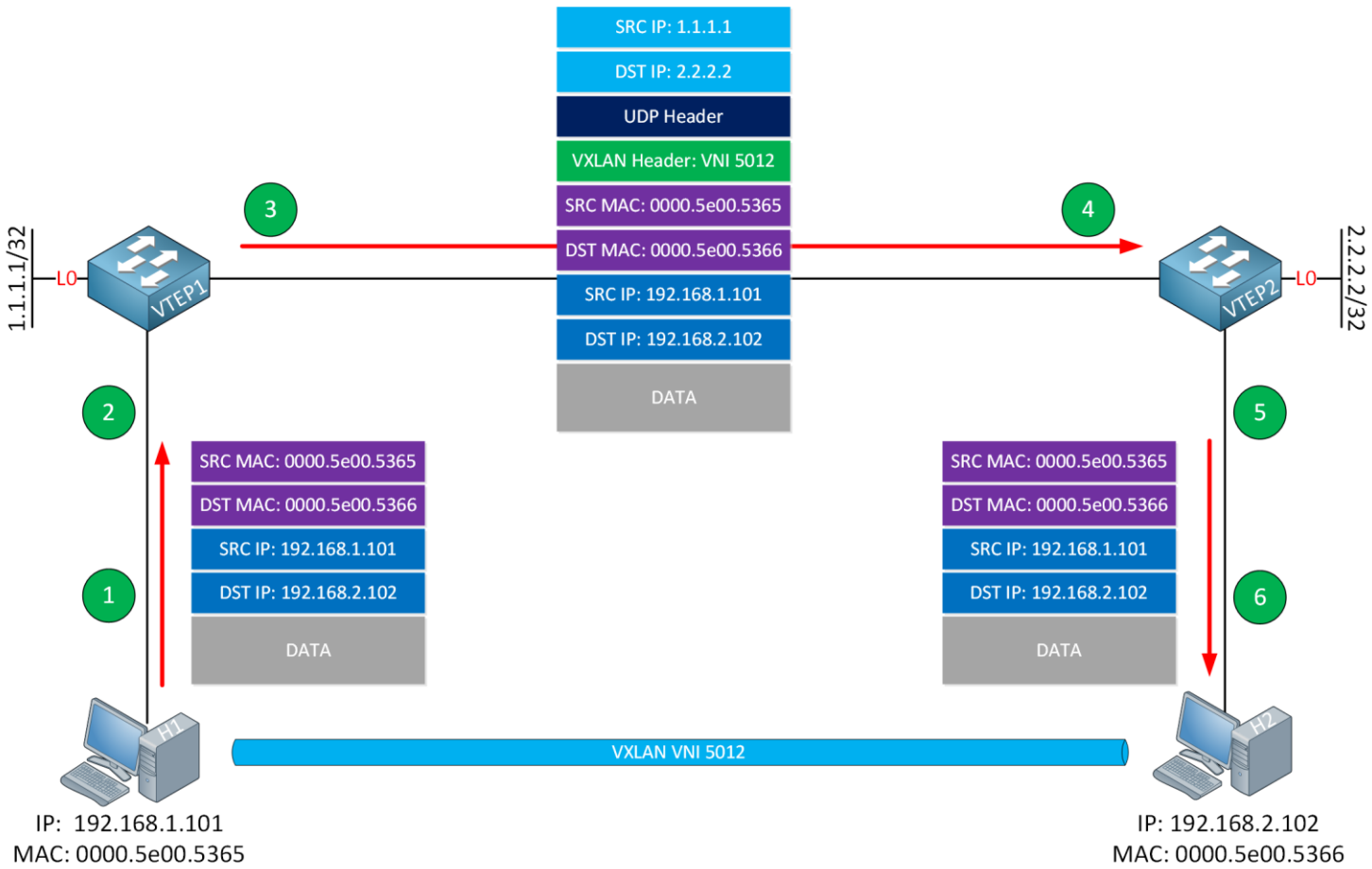
- **Overlay Network**
  The virtualized VLAN connection between 2 Layer-2 switches
- **Underlay Network**
  The physical network connection moving data from one-switch to another



Virtual Extensible LAN (VXLAN)

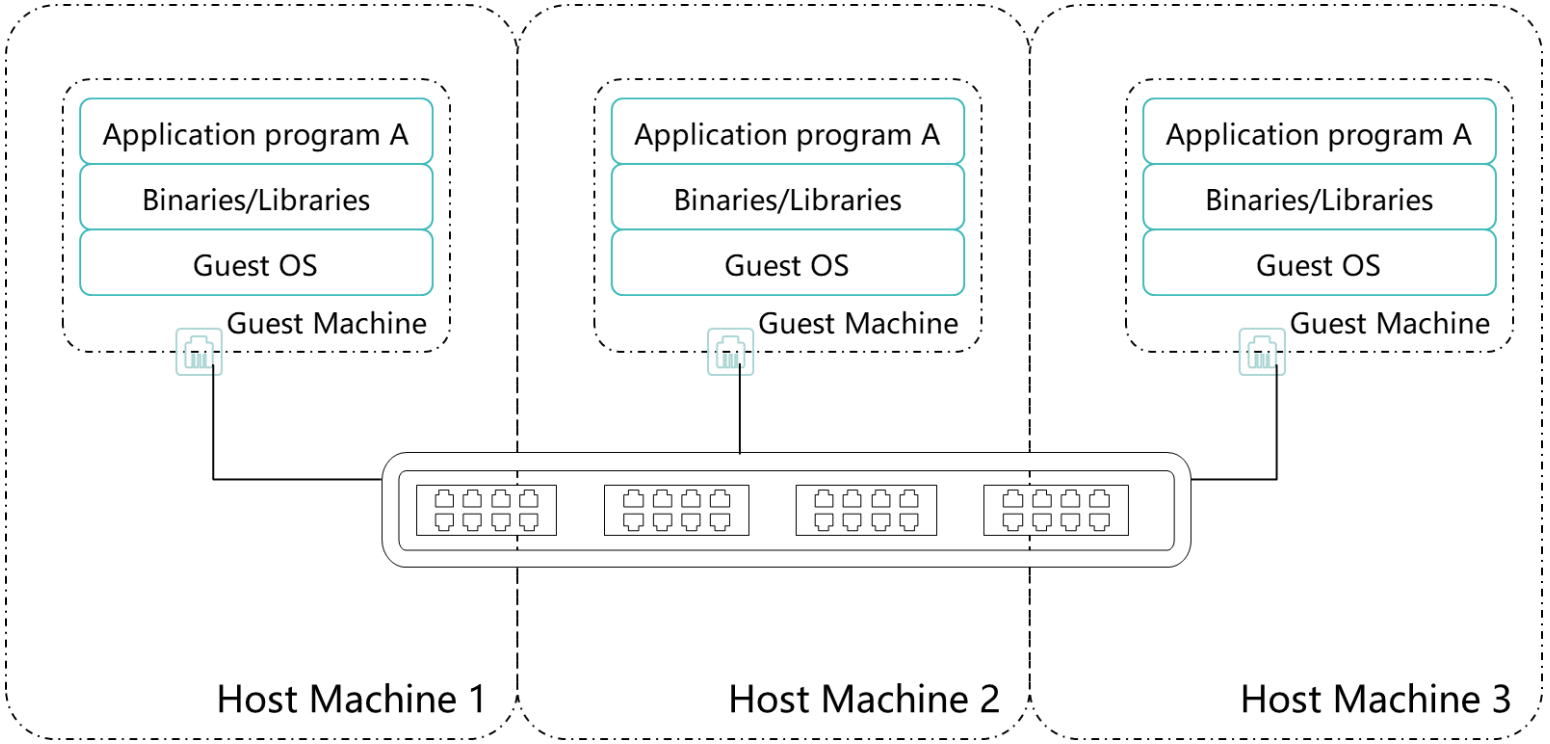
Virtualize VLAN using Layer-3 Switches/Routers

- UDP message encapsulates the VXLAN src/dst
- VXLAN header identify servers in same group
- Layer-3 devices apply routing protocols to send messages to destination



Distributed Virtual Switch (DVS)

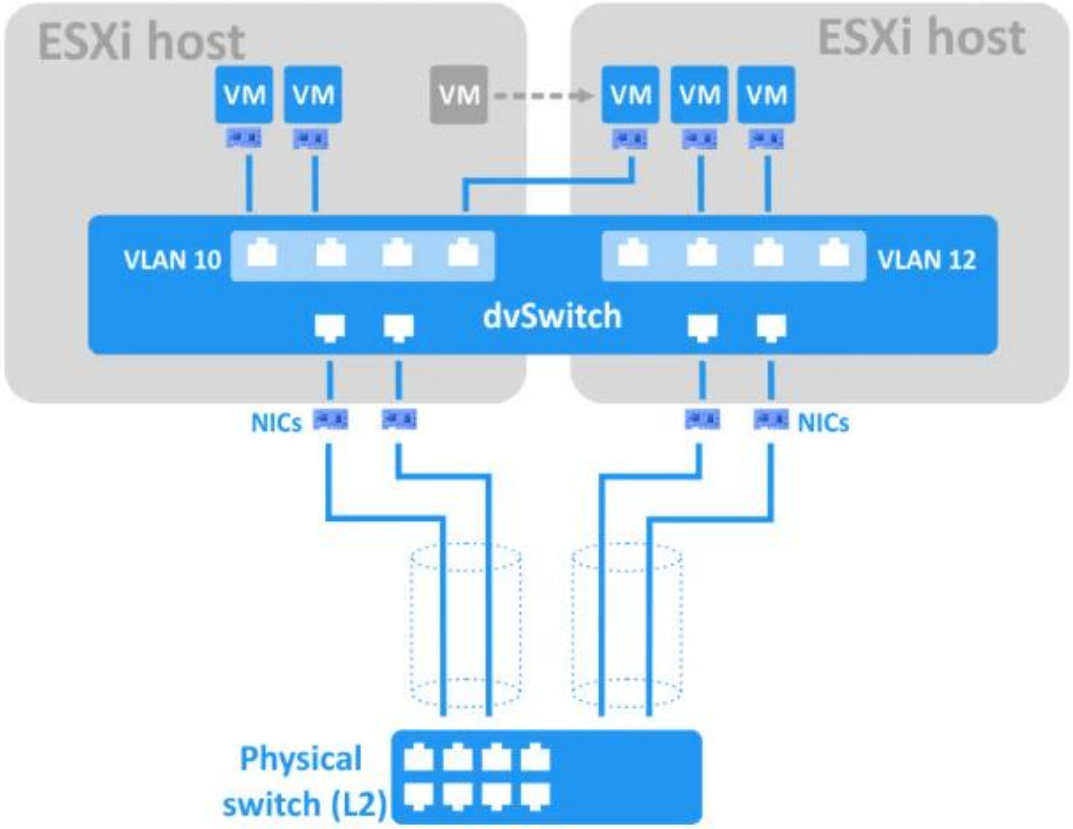
Multiple Network Hypervisors emulate the operation of a switch



Distributed Virtual Switch (DVS)

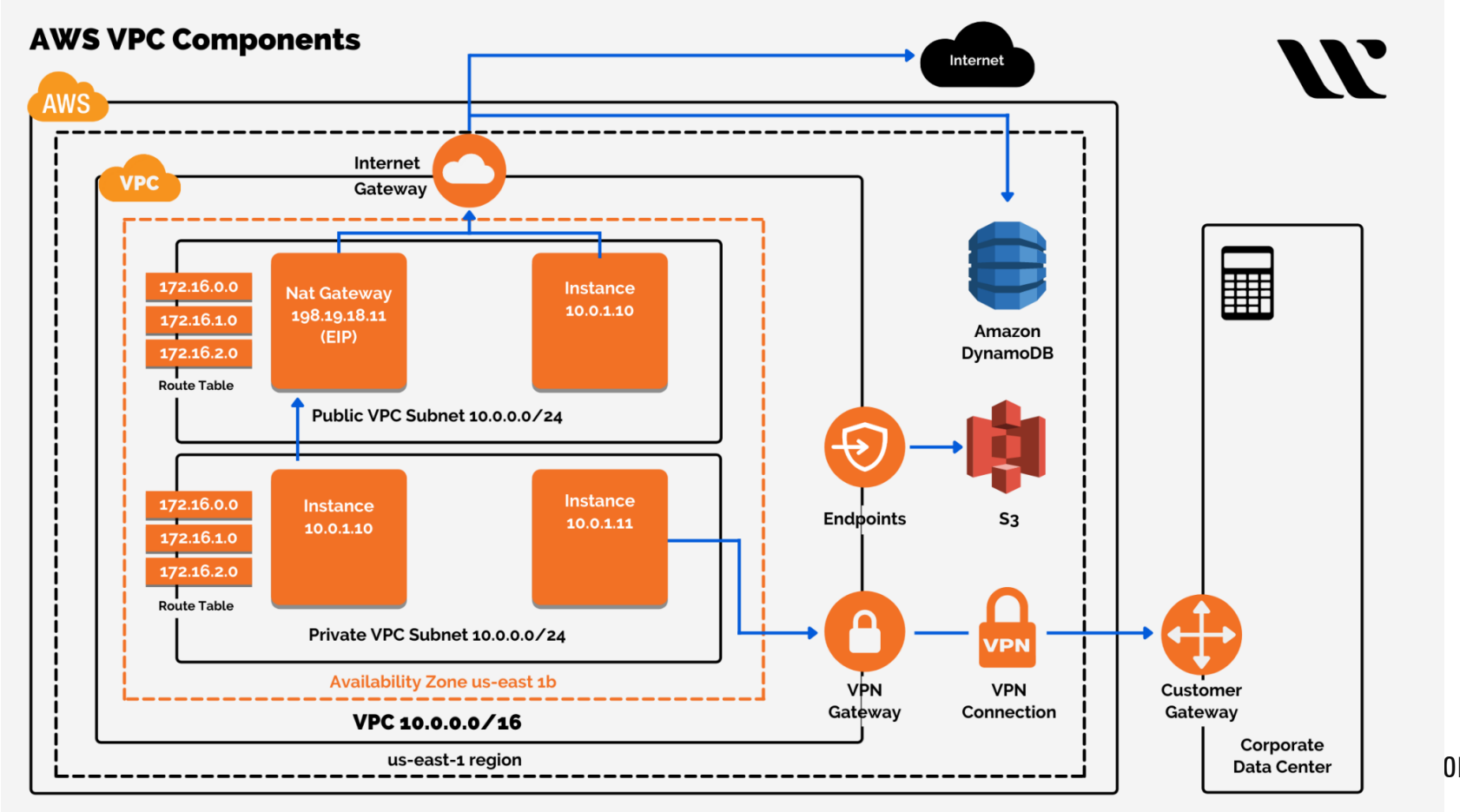
Multiple Network Hypervisors emulate the operation of a switch

- Pros
  - Provide virtual switch ports
  - Virtual switch is configurable (i.e.: control data rate)
  - LAN Ports with same attributes can form a VLAN



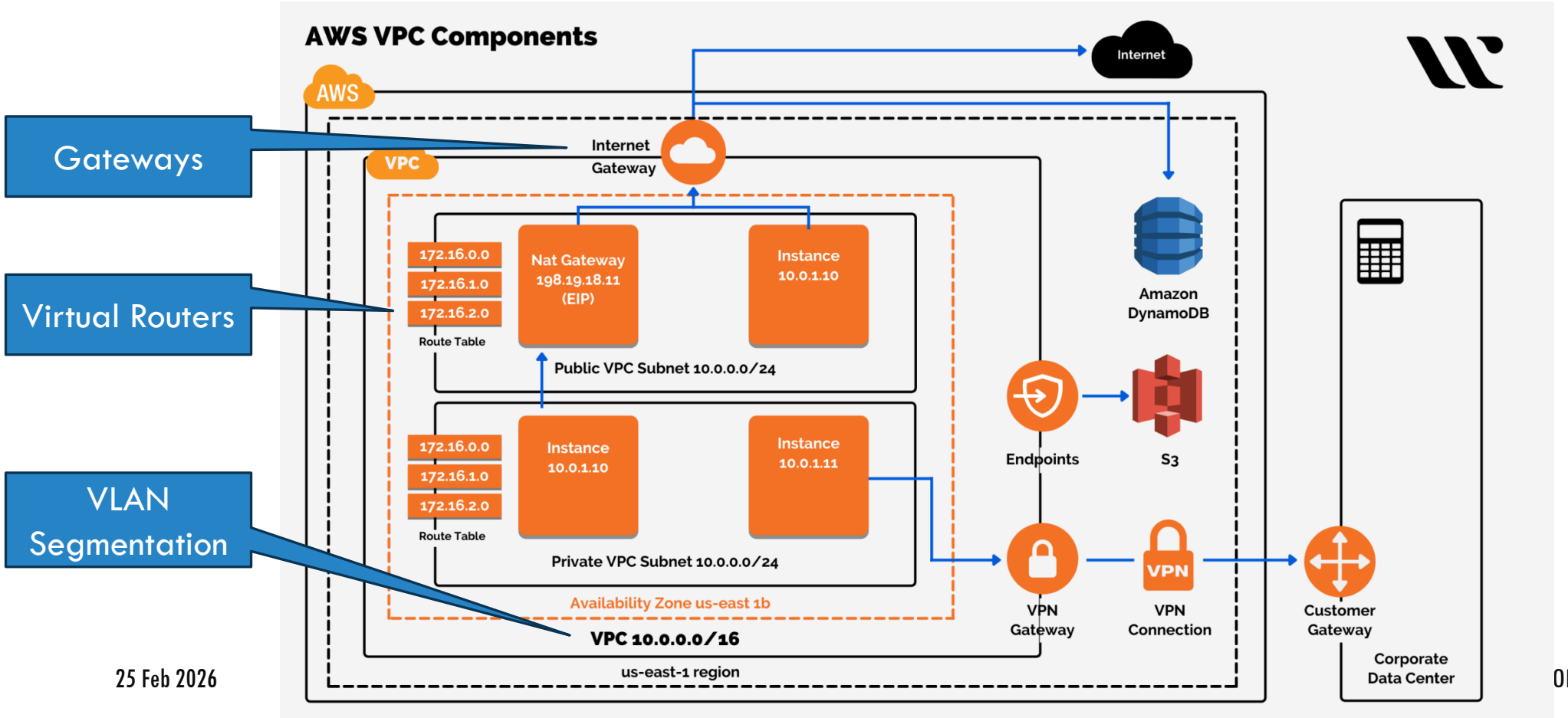
Virtual Private Clouds (VPCs)

Create private virtual networks inside public cloud



Virtual Private Clouds (VPCs)

Create private virtual networks inside public cloud



Virtual Private Clouds (VPCs)

Create private virtual networks inside public cloud

- **Pros**
  - Traffic Control (Gateways)
  - Security & Isolation (VLAN)
  - Flexible Network as a Service (Virtual Routers & subnets)

Resources

- Reference Book: Chapter 7
- https://www.geeksforgeeks.org/computer-networks/basics-computer-networking/

Thank you